

Reviewed Preprint

v1 • May 9, 2025

Not revised

Reviewed Preprint

v2 • February 17, 2026

Revised by authors

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Competing interests:

No competing interests declared

Funding:

See page 31

Reviewing editor:

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Mitochondrial ETF insufficiency drives neoplastic growth by selectively optimizing cancer bioenergetics

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eLife Assessment

The authors present an **important** set of data implicating ETFDH as an epigenetically suppressed gene in cancer with tumor suppressive functions. The evidence is **convincing**, with the authors demonstrating that suppression of ETFDH activity results in accumulation of amino acids that impact metabolism via hyperactive mTORC1.

<https://doi.org/10.7554/eLife.106587.2.sa3>

Abstract

Mitochondrial electron transport flavoprotein (ETF) insufficiency causes metabolic diseases known as a multiple acyl-CoA dehydrogenase deficiency (MADD). In contrast to muscle, ETFDH is a non-essential gene in acute lymphoblastic leukemia NALM-6 cells, and its expression is reduced across human cancers. ETF insufficiency caused by decreased ETFDH expression limits flexibility of OXPHOS fuel utilization but paradoxically increases cancer cell bioenergetics and accelerates neoplastic growth by activation of the mTORC1/BCL-6/4E-BP1 axis. Collectively, these findings reveal that while ETF insufficiency is rare and has detrimental effects in non-malignant tissues, it is common in neoplasia, where ETFDH downregulation leads to bioenergetic and signaling reprogramming that accelerate neoplastic growth.

Main

The mitochondrial electron transport chain (ETC) transfers electrons from reducing equivalents (NADH and FADH₂) towards complexes I/II, or from peripheral metabolic enzymes that reduce coenzyme Q (ubiquinone, Q) to ubiquinol (QH₂)¹. The electron transfer flavoproteins (ETFA and ETFB) accept electrons generated by the catabolism of branched-chain amino acids and fatty acids^{2–4}. ETFs interact with ETF dehydrogenase (ETFDH or ETF-ubiquinone oxidoreductase)⁵, which controls electron flow towards complex III (Supplementary Fig.1A [↗](#)). Loss-of-function mutations in ETF/ETFDH occur in metabolic diseases known as multiple acyl-CoA dehydrogenase deficiency (MADD)^{6,7}. ETFDH is also essential for complex III activity in skeletal muscle⁸. Surprisingly, we observed that in contrast to muscle, ETFDH is one of the most non-essential metabolic genes in cancer cells. Paradoxically, ETFDH downregulation leads to enhanced mitochondrial biogenesis and increased bioenergetic capacity of cancer cells. ETF insufficiency resulting from reduction in ETFDH levels triggers a tradeoff favored by cancer cells, whereby reduced flexibility in fuel utilization for oxidative phosphorylation is offset by signaling from mitochondria to mTORC1 that remodels cancer cell bioenergetics to augment neoplastic growth.

ETFDH abrogation accelerates tumor growth

ETFDH is essential in skeletal muscle cells⁸. Intriguingly, a CRISPR-Cas9 knock out (KO) screen performed in a human acute lymphoblastic leukemia cell line (NALM6)⁹ ranked *ETFDH* as a top non-essential metabolic gene that clustered with well-established tumor suppressors *TP53* and *RB1* (Supplementary Fig.1B [↗](#)). Mining DepMap also revealed that *ETFDH* is essential in only 1 out of 1150 cancer cell lines¹⁰. This motivated us to examine the role of ETFDH in cancer. *ETFDH* mRNA levels are reduced in patient samples across different cancer types, including colon (Supplementary Fig.1C [↗](#)) and breast cancer (Supplementary Fig.1D [↗](#)). Immunohistochemistry (IHC) staining confirmed that ETFDH protein levels are significantly reduced in colorectal cancer vs. non-adjacent patient-derived tissues (NAT) (Fig.1A [↗](#)).

We next abrogated the *ETFDH* gene using a CRISPR-Cas9 approach in a variety of cancer cell lines including HCT-116 (human colorectal cancer), NT2197 (murine HER2+ breast cancer), 4T1 (murine triple negative breast cancer cells), and NALM6 cells (Supplementary Fig.1E-H [↗](#)). We also investigated the effects of ETFDH loss in non-transformed normal murine mammary gland (NMuMG) cells (Supplementary Fig.1I [↗](#)), a parental immortalized mammary epithelial cell line that was transformed with an oncogenic ErbB2 variant to generate NT2197 cells¹¹. Notwithstanding their high baseline proliferation rates, ETFDH ablation significantly increased proliferation across all tested cancer cell lines (Supplementary Fig.1E-H [↗](#)). This was not caused by the inadvertent effects of the CRISPR-Cas9, as rescuing ETFDH expression in HCT-116 and NT2197 cells reverted proliferation rates to levels comparable to ETFDH-proficient controls (Supplementary Fig.1E-F [↗](#)). In contrast, ETFDH abrogation did not affect proliferation of non-transformed NMuMG cells (Supplementary Fig.1I [↗](#)). ETFDH KO HCT-116 cells also formed more colonies in soft agar (surrogate measurement of cellular neoplastic potential¹²) as compared to empty vector infected controls (WT EV), and ETFDH KO cells in which ETFDH was re-expressed (ETFDH Rescue) (Supplementary Fig.1J [↗](#)).

Based on these findings, we tested the impact of ETFDH loss on neoplastic growth *in vivo*. WT EV, ETFDH KO, or ETFDH Rescue HCT-116 cells were injected orthotopically in the caeca of SCID-BEIGE mice. Loss of ETFDH significantly increased the tumor growth rate and volume at endpoint relative to control and ETFDH Rescue cells (Fig.1B-C [↗](#)). ETFDH loss also increased NT2197 tumor growth and the final tumor volume as compared to ETFDH-proficient tumors following orthotopic mammary fat pad injections in SCID-BEIGE mice (Fig.1D-E [↗](#)). Altogether, these results show that *ETFDH* is a non-essential gene in cancer that is under-expressed across different cancer types, and that its disruption enhances neoplastic growth, while not affecting proliferation of non-transformed cells.

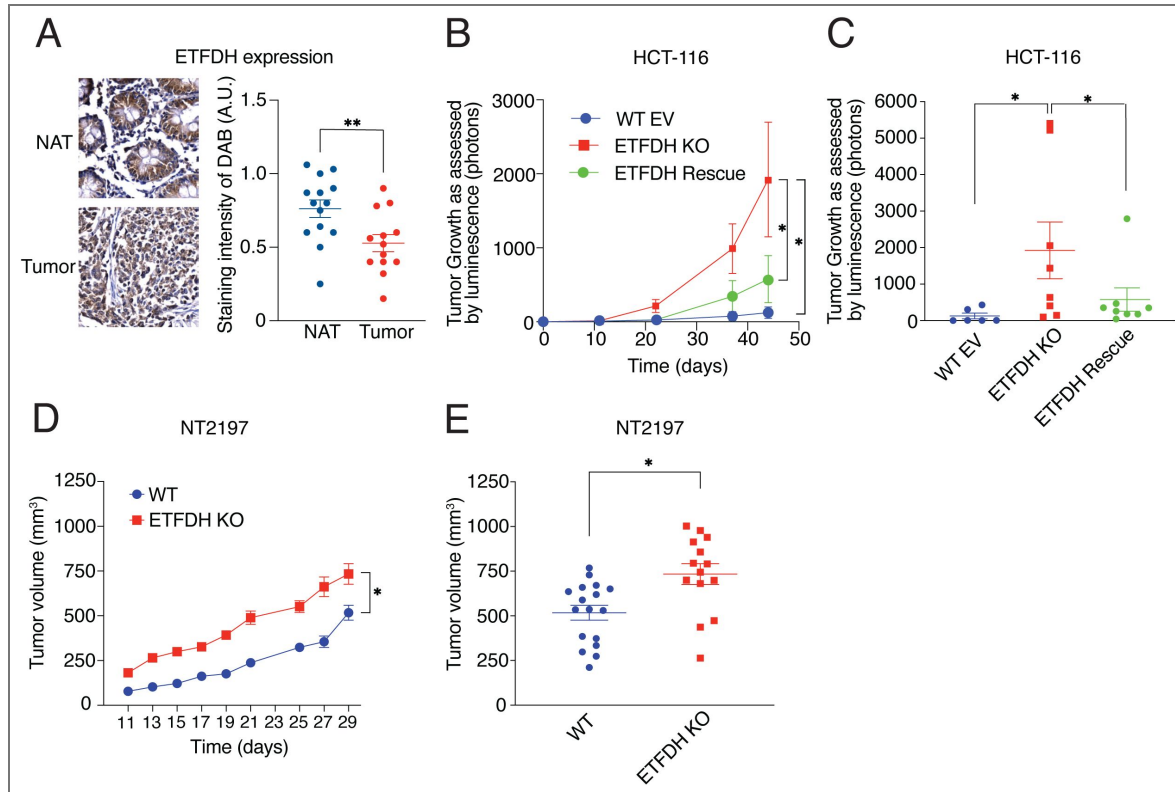


Figure 1. ETFDH loss promotes tumor growth

A) ETFDH protein levels in non-adjacent tissue (NAT) or grade 3 cancerous tissue (Tumor) were determined by immunocytochemistry (IHC) of colorectal cancer tissue microarray. Staining intensity of DAB in arbitrary units (A.U) is shown. Data are presented as means $\pm$ SEM (NAT n=14, Tumor n=13), **p<0.01, unpaired Student's *t* test. B-C) Tumor growth of WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells following intra-caecal injection (B) and endpoint tumor volumes (C). Growth was assessed by luminescence (photons). Data are presented as means $\pm$ SEM (WT EV n=6, ETFDH KO n=8, ETFDH Rescue n=8), *p<0.05, one-way ANOVA, Tukey's post-hoc test. D-E) Tumor growth of WT and ETFDH KO NT2197 cells following mammary fat-pad injection (D) and endpoint tumor volumes (E). Growth was assessed using a caliper. Data are presented as means $\pm$ SEM (WT n=17, ETFDH KO n=17), *p<0.05, unpaired Student's *t* test.

ETFDH abrogation results in ETF insufficiency that paradoxically increases bioenergetic capacity of cancer cells

In non-cancerous cells, ETFDH transfers electrons from catabolism of fatty acids and amino acids to the ETC (Supplementary Fig.2A [\[link\]](#)). Palmitate oxidation (monitored by oxygen consumption) and ^{13}C leucine labelling into citrate (citrate $m+2$) were strongly attenuated by ETFDH abrogation in HCT-116 cells (Supplementary Fig.2B-C [\[link\]](#)), which confirmed that ETF plays a role in fatty and amino acid catabolism in malignant cells. Thus, ETFDH loss induces ETF insufficiency in cancer cells that mirrors defects observed in non-malignant tissues. Surprisingly, ETFDH loss in HCT-116 cells increased oxygen consumption and extracellular acidification, relative to control WT EV and ETFDH Rescues (Fig.2A-B [\[link\]](#)). ETFDH KO HCT-116 cells produced higher levels of ATP from oxidative phosphorylation (J ATP ox) as compared to ETFDH-proficient cells (Fig.2C-D [\[link\]](#)). Consequently, ETFDH loss increased the bioenergetic capacity of HCT-116 cells (Fig.2E [\[link\]](#)). Comparable results were observed in NT2197 cells (Supplementary Fig.2D-H [\[link\]](#)). Collectively, these findings suggest that although ETF insufficiency limits the ability of cancer cells to utilize lipids and amino acids, it paradoxically upregulates their mitochondrial metabolism and bioenergetic capacity.

Glutamine plays an important role in fueling mitochondrial metabolism and *de novo* nucleotide biosynthesis to support cancer cell proliferation¹³. ETFDH abrogation increased glutamine uptake and glutamate production in both HCT-116 (Fig.2F [\[link\]](#)) and NT2197 (Supplementary Fig.2I [\[link\]](#)) cells. ETFDH loss increased ^{13}C -glutamine incorporation in the forward direction (green) and the reverse direction (purple) of the citric acid cycle (CAC; Fig.2G-H [\[link\]](#), Supplementary Fig.2J [\[link\]](#)), suggesting higher glutamine tracing into the CAC, as well as increased reductive glutamine metabolism. Indeed, the steady state levels of several nucleotides were also increased following ETFDH loss (Supplementary Fig.2K [\[link\]](#)). To highlight the dependence of glutamine in contributing to the proliferation of cells upon ETFDH abrogation, HCT-116 WT EV and ETFDH KO cells were grown in the absence or presence of glutamine. Accordingly, ETFDH KO HCT-116 cells exhibited greater sensitivity to glutamine deprivation than control WT EV cells (Fig.2I [\[link\]](#)). Collectively, these data demonstrate that ETF insufficiency in cancer cells remodels mitochondrial metabolism and increases the glutamine consumption and anaplerosis.

ETFDH loss induces intracellular accumulation of amino acids, mTOR signaling and protein synthesis

Consistent with the established role of ETFDH in amino acid catabolism², and reduction of leucine consumption upon ETFDH abrogation (Supplementary Fig.2C [\[link\]](#)), steady state levels of most intracellular amino acids were increased in ETFDH KO vs. WT HCT-116 and NT2197 cells (Fig.3A [\[link\]](#)). Moreover, ETFDH-deficient HCT-116 cells exhibited higher rates of protein synthesis relative to ETFDH-proficient cells (Fig.3B [\[link\]](#)). Amino acids activate the mechanistic target of rapamycin (mTOR) which stimulates protein synthesis¹⁴. Indeed, mTORC1 signaling was higher in ETFDH-deficient than ETFDH-proficient HCT-116 and NT2197 cells as illustrated by increased phosphorylation of ribosomal protein S6 kinases (S6K1/2; Thr389) and their substrate ribosomal protein S6 (rpS6; Ser240/244) (Fig.3C [\[link\]](#)). Re-expression of ETFDH in ETFDH KO cells resulted in normalization of mTORC1 signaling to the levels observed in WT EV cells (Fig.3D [\[link\]](#)). mTORC2 signaling was also augmented upon ETFDH abrogation in HCT-116 and NALM6 cells as evidenced by increased AKT phosphorylation (Ser473) (Supplementary Fig.3A-B [\[link\]](#)). Despite bioenergetic rewiring, ETFDH loss did not exert a major effect on the activity of adenosine monophosphate-activated protein kinase (AMPK) as monitored by its phosphorylation (Thr172) or phosphorylation of its substrate acetyl-CoA carboxylase (ACC at S79) (Supplementary Fig.3A-B [\[link\]](#)).

Consistent with mTOR hyperactivation, ETFDH KO HCT-116 cells were more sensitive to the active site mTOR inhibitor Torin1¹⁵ as compared to their ETFDH-proficient counterparts (Supplementary Fig.3C [\[link\]](#)). To further determine which mTOR complex is responsible for mediating the hyper-proliferative phenotype triggered by ETFDH loss, we selectively inactivated mTORC1 or mTORC2 by depleting RAPTOR or RICTOR, respectively using small hairpin RNA (shRNA) in WT and ETFDH

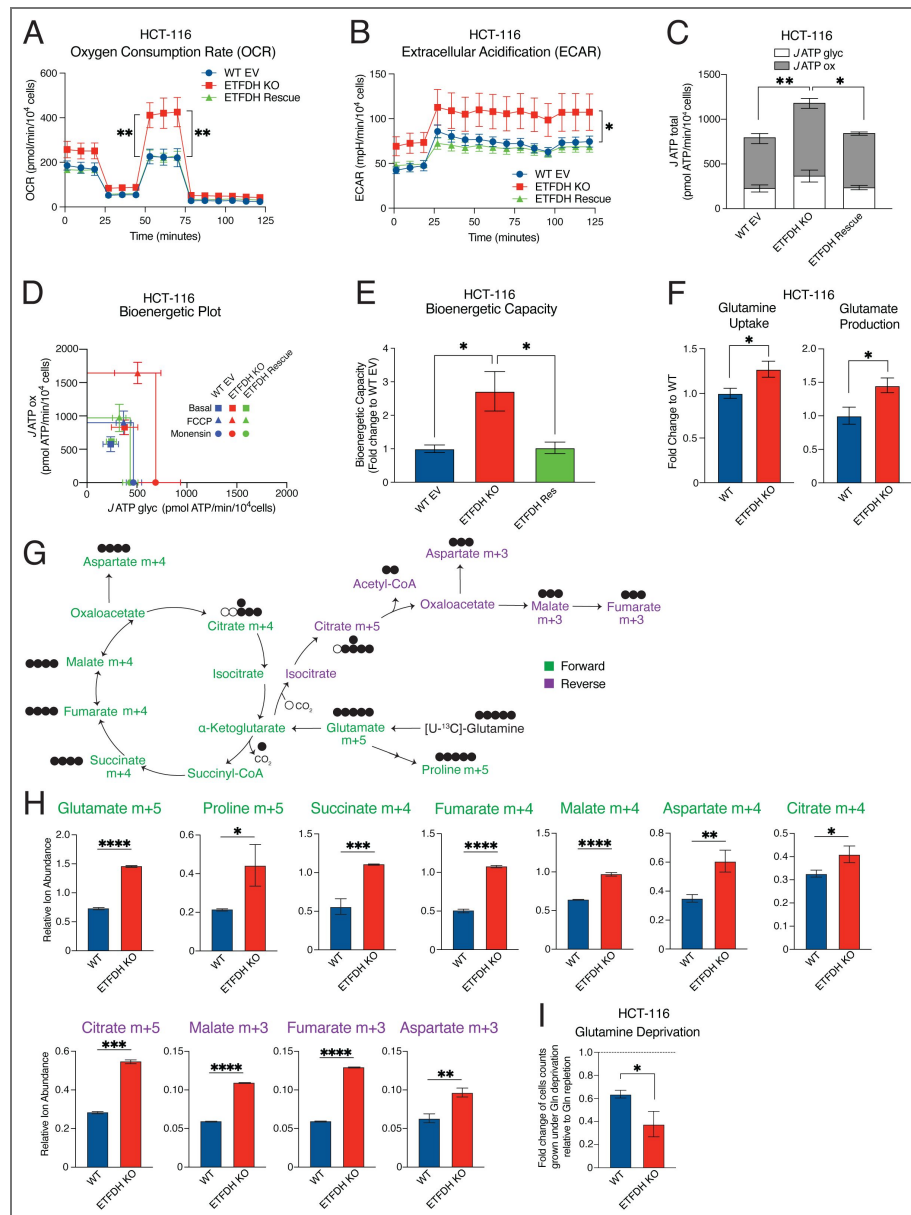


Figure 2. Absence of ETFDH promotes mitochondrial metabolism

A-B) Oxygen consumption (A) and extracellular acidification (B) in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells were determined using a Seahorse bioanalyzer. Data are normalized to cell count and presented as means $\pm$ SD ($n=4$), * $p<0.05$, ** $p<0.01$, one-way ANOVA, Tukey's post-hoc test. C) Basal J ATP calculations from WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells. J ATP ox represents ATP production from oxidative phosphorylation, while J ATP glycolysis is ATP production from glycolysis. Comparison between J ATP ox (grey bars; top) and J ATP glycolysis (white bars; bottom) is shown. Data are presented as means $\pm$ SD ($n=4$), * $p<0.05$, ** $p<0.01$, one-way ANOVA, Tukey's post-hoc test. D-E) Bioenergetic plot for Basal, FCCP, and Monensin J ATP fluxes (D) and bioenergetic capacity (E) of WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells. Data are presented as means $\pm$ SD ($n=4$), * $p<0.05$, one-way ANOVA, Tukey's post-hoc test. F) Glutamine uptake and glutamate production in WT and ETFDH KO HCT-116 cells. Results are shown as fold changes of ETFDH KO cells relative to WT cells. Data are presented as means $\pm$ SD ($n=3$), * $p<0.05$, paired Student's t test. G) Schematic of ^{13}C -glutamine tracing throughout the citric acid cycle (CAC). Isotopomers labelled in green depict ^{13}C -glutamine tracing in the forward direction of the CAC, while those in purple represent reverse tracing. H) Relative abundance of ^{13}C -labelled metabolites in the forward (green) and reverse (purple) directions from WT (blue) and ETFDH KO (red) NT2197 cells. Data are presented as means $\pm$ SD ($n=3$), * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$, paired Student's t test. I) Proliferation of WT and ETFDH KO HCT-116 cells grown in the absence or presence of glutamine for 48 hours. Results are shown as fold changes of cell counts under glutamine deprivation relative to those under glutamine repleted conditions. Data are presented as means $\pm$ SD ($n=3$), * $p<0.05$, paired Student's t test.

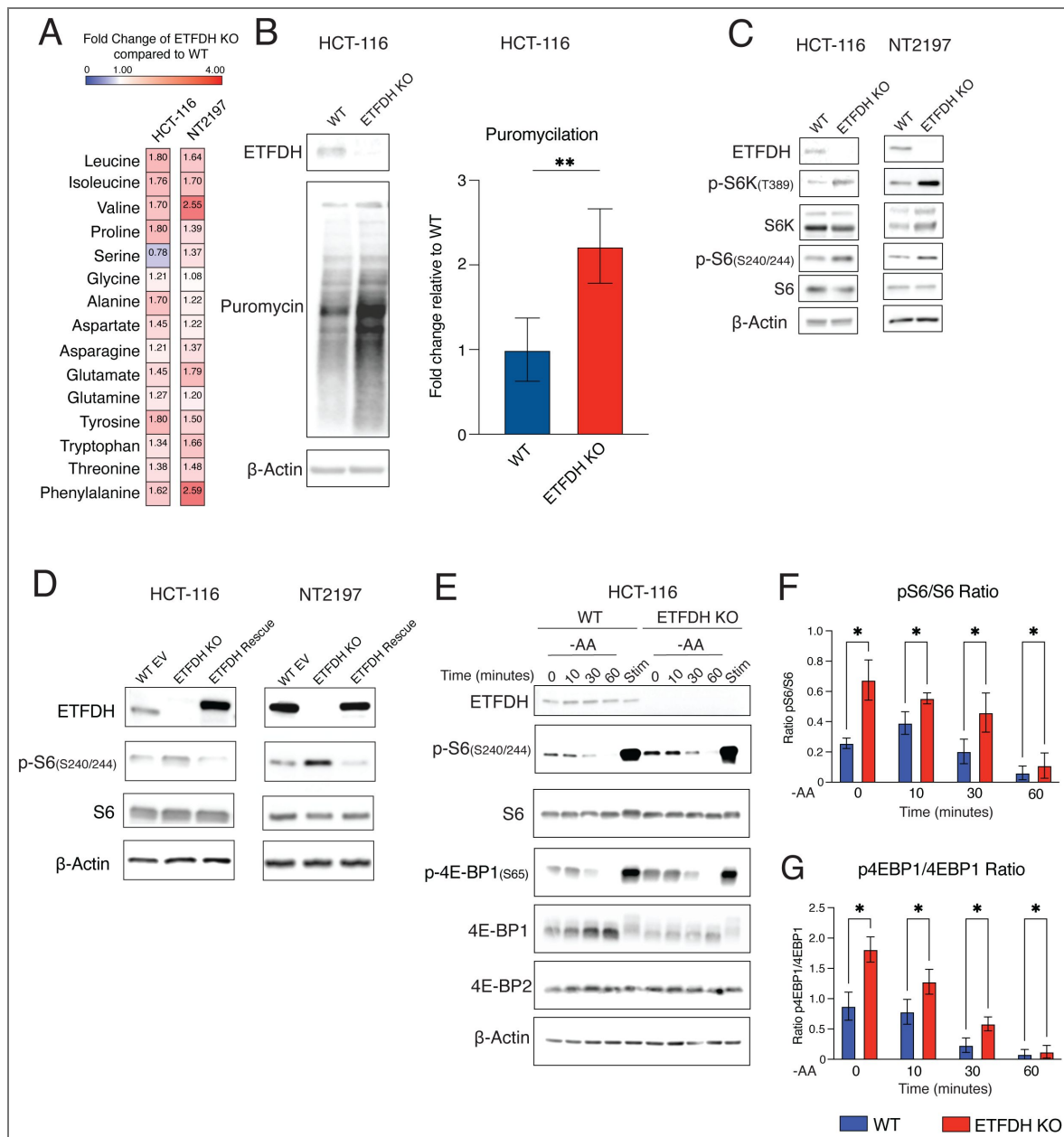


Figure 3. ETFDH loss bolsters mTORC1 signaling and downregulates 4E-BP1 protein levels.

A) Amino acid profile in WT and ETFDH KO HCT-116 and NT2197 cells. Data are presented as fold change in amino acid levels in ETFDH KO compared to WT cells (n=3). B) Puromycin incorporation assays in WT and ETFDH KO HCT-116 cells. Levels and phosphorylation status of indicated proteins were determined via western blotting using indicated antibodies. β -Actin was used as a loading control (representative blots of n=3). Quantification of puromycin incorporation is presented as fold change in ETFDH KO vs. WT cells. C) Levels and phosphorylation status of indicated proteins in WT or ETFDH KO HCT-116 and NT2197 cells were determined by western blotting. β -Actin served as a loading control (representative blots of n=3). D) Levels and phosphorylation status of indicated proteins in WT EV, ETFDH KO, or ETFDH Rescue HCT-116 and NT2197 cells were determined by western blotting. β -Actin was used as a loading control (representative blots of n=3). E) Total and phosphoprotein levels in WT or ETFDH KO HCT-116 cells were determined by western blotting using indicated proteins. Cells were serum starved overnight, then depleted of amino acids for indicated timepoints (0, 10, 30, 60 minutes), or stimulated with FBS for 4 hours (Stim). β -Actin was used as loading control (representative blots of n=3). F) Quantification of pS6(S240/244)/S6 ratio from data presented in panel E. Data are presented as means $\pm$ SD (n=3), *p<0.05, one-way ANOVA, Tukey's post-hoc test. G) Quantification of p4E-BP1(S65)/4E-BP1 ratio from data presented in panel E. Data are presented as means $\pm$ SD (n=3), *p<0.05, one-way ANOVA, Tukey's post-hoc test.

KO HCT-116 cells (Supplementary Fig.3D [↗](#)). Depletion of RAPTOR, but not RICTOR strongly attenuated the proliferation of ETFDH KO cells (Supplementary Fig.3E-F [↗](#)). Moreover, the anti-proliferative effects of bisteric mTORC1-specific inhibitor Bis-35x¹⁶ were stronger in ETFDH KO vs. ETFDH Rescue HCT-116 cells (Supplementary Fig.3G [↗](#)). Altogether, these data suggest that mTORC1, but not mTORC2 acts as a major mediator of hyper-proliferative phenotype triggered by ETFDH loss in cancer cells. Accordingly, the effects of amino acid deprivation on phospho-rpS6 (Ser240/244), and phospho-4E-BP1 (Ser65) were attenuated in ETFDH KO vs. WT control cells, which is consistent with the intracellular accumulation of amino acids upon ETFDH disruption (Fig.3E-G [↗](#)). In turn, serum stimulation (Stim) resulted in comparable induction in phosphorylation of rpS6 (Ser240/244) and 4E-BP1 (Ser65) in ETFDH-deficient and proficient cells (Fig.3E [↗](#)). Collectively, these findings suggest that ETFDH loss induces mTORC1 signaling at least in part by increasing intracellular amino acid levels.

Amino acid deprivation leads to general control nonderepressible 2 (GCN2)-dependent phosphorylation of the alpha subunit of eIF2 and subsequent induction of activating transcription factor 4 (ATF4) synthesis during the integrated stress response (ISR)¹⁷. Consistent with an increase in intracellular amino acid, phospho-eIF2 α (S51) levels appeared to be modestly reduced in ETFDH-deficient vs. proficient HCT-116 cells under both serum starvation and amino acid depletion (Supplementary Fig.3H [↗](#)). However, the ETFDH status in the cells did not affect ATF4 levels at baseline or upon amino acid depletion (Supplementary Fig.3H [↗](#)). Altogether, these data demonstrate that ETFDH loss induces mTORC1 signaling and protein synthesis, which appears to be at least in part driven by intracellular accumulation of amino acids.

ETFDH loss diminishes 4E-BP1, but not 4E-BP2 protein levels

mTORC1 positively controls cell proliferation through the eukaryotic translation initiation factor 4E-binding proteins (4E-BPs; 4E-BP1-3 in mammals)¹⁸. Of note, 4E-BP3 is expressed in a tissue-restricted manner, and does not appear to play a major role in regulating proliferation¹⁸. mTORC1 phosphorylates 4E-BPs to promote eukaryotic translation initiation factor 4F (eIF4F) complex assembly by dissociating 4E-BPs from eIF4E, thereby stimulating cap-dependent translation initiation¹⁹. Consistent with the increased mTORC1 activity, ETFDH-deficient cells exhibited increased phosphorylation of 4E-BP1 (Ser65), which was rescued upon re-expression of ETFDH in HCT-116 and NT2197 cells (Fig.4A [↗](#)). Unexpectedly, we observed that ETFDH loss coincided with a dramatic reduction in total 4E-BP1 protein levels in HCT-116 (Fig.4A [↗](#)), NT2197 (Fig.4A [↗](#)), NALM6 (Supplementary Fig.4A), and 4T1 cells (Supplementary Fig.4B). We also treated WT and ETFDH KO HCT-116 cell lysates with λ -phosphatase to exclude potential confounding effects of different 4E-BP1 phosphorylation states, which confirmed that disruption of ETFDH is paralleled by reduction in 4E-BP1 levels (Fig.4B [↗](#)). Notably, loss of ETFDH did not exert a major effect on 4E-BP2 protein abundance in HCT-116 (Fig.4A [↗](#)), NT2197 (Fig.4A [↗](#)), or 4T1 cells (Supplementary Fig.4B [↗](#)). In addition, 4E-BP1, but not 4E-BP2 protein levels were reduced in ETFDH-deficient HCT-116 as compared to WT tumors (Fig.4C-E [↗](#)). Strikingly, loss of ETFDH in non-transformed NMuMG cells did not exert a major effect on mTORC1 signaling (Supplementary Fig.4C) and failed to alter 4E-BP1 protein levels (Supplementary Fig.4C). Collectively, these findings indicate that ETF insufficiency is paralleled by a selective decrease in 4E-BP1 protein levels in cancer, but not non-transformed cells.

Decrease in 4E-BP1 protein levels underpins the bioenergetic rewiring induced by ETF insufficiency in cancer cells

Considering that in mammals 4E-BPs play a major role in mediating the effects of mTORC1 on proliferation, neoplastic growth, mitochondrial functions and protein synthesis^{18,20,21}, we next investigated whether the observed decrease in 4E-BP1 levels contributes to increased proliferation and tumor growth caused by ETFDH abrogation. To this end, we overexpressed 4E-BP1 protein in ETFDH-deficient NT2197 cells by infecting them with a retrovirus wherein *EIF4EBP1* expression was driven by an exogenous promoter¹⁸. Importantly, 4E-BP1 overexpression in ETFDH KO NT2197 cells decreased eIF4F complex assembly, as evidenced by m⁷-GTP pulldown (Fig.4F [↗](#)). 4E-

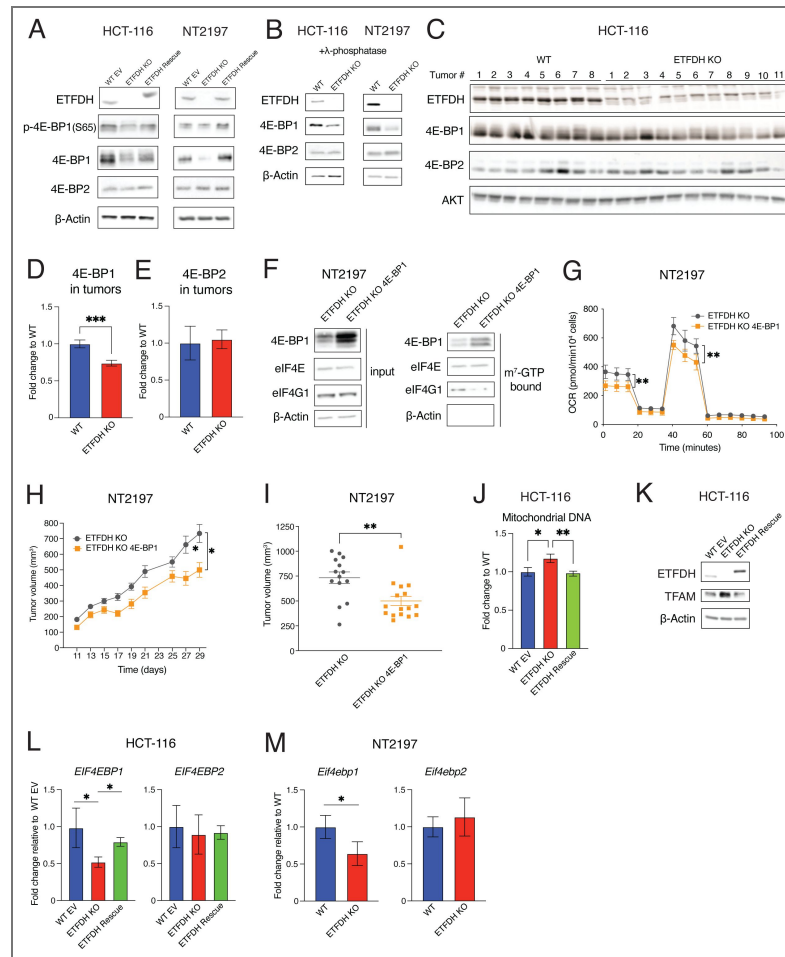


Figure 4. Repression of *EIF4EBP1* transcription mediates the effects of ETFDH loss on cancer bioenergetics and tumor growth

A) Levels and phosphorylation status of indicated proteins in WT EV, ETFDH KO, or ETFDH Rescue HCT-116 and NT2197 cells were determined by western blotting. β -Actin was used as a loading control (representative blots of $n=3$). B) Levels of indicated protein in λ -phosphatase treated WT or ETFDH KO HCT-116 and NT2197 cell lysates were monitored by western blotting. β -Actin was used as a loading control (representative blots of $n=3$). C) Indicated protein levels in WT or ETFDH KO HCT-116 tumors were determined by western blotting. AKT was used as loading control (WT $n=8$, ETFDH KO $n=11$). D-E) Quantification of 4E-BP1 (D) and 4E-BP2 (E) from tumors described in panel C. Data are presented as means $\pm$ SEM (WT $n=8$, ETFDH KO $n=11$), *** $p<0.001$, unpaired Student's t test. F) m^7 GDP pulldown assay in ETFDH KO and ETFDH KO 4E-BP1 NT2197 cells. Specified protein levels in m^7 GDP pulldown or input were determined by western blot. β -Actin was used as loading control (input) and to exclude contamination in the pulldown material (m^7 GTP bound) (representative blots of $n=3$). G) Oxygen consumption of ETFDH KO and ETFDH KO 4E-BP1 NT2197 cells. Data are normalized to cell count and presented as means $\pm$ SD ($n=5$), ** $p<0.01$, Student's t test. H-I) Tumor growth of ETFDH KO and ETFDH KO 4E-BP1 NT2197 cells following mammary fat-pad injection (H) and endpoint tumor volumes (I). Growth was assessed using caliper measurements. ETFDH KO measurements are the same as in Fig.1D, as these growth curves were obtained in the same experiment. Data are presented as means $\pm$ SEM (ETFDH KO $n=17$, ETFDH KO 4E-BP1 $n=18$), ** $p<0.01$, unpaired Student's t test. J) Mitochondrial DNA in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells was quantified by qPCR. Mitochondrial DNA (mtDNA) content was normalized to genomic DNA (gDNA). Data are presented as fold change relative to WT EV cells $\pm$ SD ($n=3$), * $p<0.05$, ** $p<0.01$, one-way ANOVA, Tukey's post-hoc test. K) Levels of indicated proteins in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells were assessed by western blotting. β -Actin was used as a loading control (representative blots of $n=3$). L) *EIF4EBP1* and *EIF4EBP2* mRNA levels in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells were determined by RT-qPCR. *PP1A* was used as a house-keeping gene. Data are presented as fold change of *EIF4EBP1/PP1A* and *EIF4EBP2/PP1A* ratios relative to WT EV cells (set to 1) $\pm$ SD ($n=3$), * $p<0.05$, one-way ANOVA, Tukey's post-hoc test. M) *Eif4ebp1* and *Eif4ebp2* mRNA abundance in WT and ETFDH NT2197 KO cells was determined by RT-qPCR. *Actb* was used as a house-keeping gene. Data are presented as fold change in *Eif4ebp1/Actb* and *Eif4ebp2/Actb* ratios relative to WT cells $\pm$ SD ($n=3$), * $p<0.05$, paired Student's t test.

BP1 overexpression also reduced oxygen consumption in ETFDH KO NT2197 cells, relative to empty vector infected cells (Fig.4G [↗](#)). Moreover, restoring 4E-BP1 protein levels decreased cell proliferation (Supplementary Fig.4D) and anchorage-independent growth (Supplementary Fig.4E) of ETFDH KO HCT-116 cells. Lastly, 4E-BP1 overexpression decreased growth and the final volume of tumors formed by NT2197 ETFDH KO cells in SCID-BEIGE mice (Fig.4H-I [↗](#)). Collectively, these data indicate that decreased 4E-BP1 protein levels mediate the pro-neoplastic effects of ETF insufficiency.

The mTORC1/4E-BP axis stimulates mitochondrial biogenesis and metabolism by altering the translation of nuclear-encoded mitochondrial mRNAs including mitochondrial transcription factor A (TFAM)^{21,22}. ETFDH loss increased mitochondrial DNA levels (Fig.4J [↗](#)) and mass (Supplementary Fig.4F [↗](#)) which was paralleled by TFAM induction (Fig.4K [↗](#)). Collectively, these data show that the loss of 4E-BP1 plays a prominent role in mediating the effects of ETF insufficiency on bioenergetic reprogramming of cancer cells.

ETFDH loss decreases *EIF4EBP1* transcription via BCL-6

We next set out to determine the level(s) at which 4E-BP1 protein levels are downregulated in ETF insufficient cancer cells. ETFDH loss did not have a major effect on 4E-BP1 protein synthesis or turnover (Supplementary Fig.4G-H [↗](#)), but it decreased the *EIF4EBP1* mRNA levels in ETFDH-deficient vs. ETFDH-proficient HCT-116 (Fig.4L [↗](#)) and NT2197 cells (Fig.4M [↗](#)). *EIF4EBP1* and *ETFDH* mRNA levels were also positively correlated in tumors isolated from colorectal adenocarcinoma patients (Supplementary Fig.4I [↗](#)). In turn, consistent with no apparent differences in 4E-BP2 protein levels, the levels of *EIF4EBP2* mRNA were unaffected by the ETFDH status in both cell lines (Fig.4L-M [↗](#)). Notably, observed decrease in *EIF4EBP1* mRNA levels was not caused by the effects of ETFDH loss on mRNA stability (Supplementary Fig.4J). Based on these data, we concluded that ETFDH abrogation is paralleled by suppression of *EIF4EBP1* transcription and sought to identify transcription factors that may mediate the effects of ETFDH abrogation on suppression of *EIF4EBP1* expression. ATF4, Snail and Slug have been shown to stimulate transcription of *EIF4EBP1*, but not *EIF4EBP2*^{23,24}. However, ATF4, Snail and Slug proteins remain unchanged upon ETFDH disruption (Supplementary Fig.3H, Supplementary Fig.5A). We also re-analyzed publicly available RNA-seq datasets, which revealed that *EIF4EBP1*, but not *EIF4EBP2* expression was increased following STAT1 abrogation in MT4788 breast cancer cells²⁵. Indeed, we observed that MT4788 STAT1 KO cells have increased 4E-BP1 protein levels relative to parental cells (Supplementary Fig.5B). Accordingly, ETFDH KO cells displayed increased STAT1 and downregulated 4E-BP1 protein levels (Supplementary Fig.5C). To determine the impact of STAT1 on *EIF4EBP1* or *EIF4EBP2* promoter occupancy, we performed chromatin immunoprecipitation qPCR (ChIP-qPCR) assay in control (WT EV), ETFDH KO, or ETFDH rescue HCT-116 cells (Supplementary Fig.5D-E). Binding of STAT1 was however comparable to IgG across all tested cell lines thereby suggesting that is unlikely that STAT1 directly suppresses *EIF4EBP1* transcription in the context of ETFDH loss (Supplementary Fig.5D-E). Furthermore, reactive oxygen species (ROS) can also influence redox-responsive transcription²⁶. Although loss of ETFDH induced ROS levels (Supplementary Fig.5F), 4E-BP1 protein abundance was not altered following treatment with the antioxidant N-acetyl-cysteine (NAC) (Supplementary Data Fig.5G).

We next investigated the potential role of B-cell lymphoma 6 (BCL-6)^{27–29}, which was previously suggested to suppress *EIF4EBP1* transcription³⁰. Notably, *ETFDH* abrogation in HCT-116 and NT2197 cells resulted in a marked increase in BCL-6 protein levels, compared to ETFDH-proficient control cells (Fig.5A [↗](#)). These changes in BCL-6 protein levels were mTOR-sensitive (Fig.5B [↗](#)) and not accompanied by the alterations in *BCL6* mRNA abundance (Fig.5C [↗](#)), which is consistent with recently reported mTOR-dependent translational activation of BCL-6^{31–33}. Indeed, *BCL6* mRNA translation was increased in HCT-116 ETFDH KO relative to WT cells (Fig.5D [↗](#)). Taken together, these data point out that ETFDH disruption induces BCL-6 protein synthesis in an mTOR-dependent manner. ETF insufficiency increased BCL-6 binding to the *EIF4EBP1*, but not the *EIF4EBP2* promoter (Fig.5E-F [↗](#)). BCL-6 silencing in NT2197 and 4T1 ETFDH KO cells increased 4E-

BP1 protein levels (Fig.5G-H). Depletion of BCL-6 also attenuated proliferation of 4T1 ETFDH KO cells (Fig.5I). Collectively, these findings show that BCL-6 mediates the effects of ETF insufficiency on reduction in *EIF4EBP1* expression and increased proliferation of cancer cells.

ETFDH acts as a haploinsufficient tumor suppressor

Data available in cBioPortal from 2683 samples across cancer types³⁴ indicate that missense or truncating mutations in ETFDH are rare (0.3%) (Fig.6A). This suggests that ETFDH is not lost, but rather downregulated in human cancers. We observed that DNA methylation inhibitors, 5-azacytidine (5-aza) and 5-aza-2'-deoxycytidine (5-aza-2-deoxyC) increased ETFDH protein levels in HCT-116 cells (Fig.6B), thus suggesting that one of the potential mechanisms of downregulated *ETFDH* expression in neoplasia may stem from increased DNA methylation. Based on these observations, we developed a model that more closely reflects ETFDH downregulation, but not a complete loss, in human tumors using a method based on targeted poly(A) track insertion within endogenous genes³⁵. The length of the poly(A) track directly correlates with the reduction in amount of protein produced, allowing for the investigation of gene-dosage effects on ensuing phenotypes^{35,36}. Accordingly, mutants were designed by inserting poly(A) tracks of 12 and 18 adenosines [12A, equivalent to 4 consecutive AAA (Lys) codons (AAA)₄; 18A, equivalent to 6 consecutive AAA (Lys) codons (AAA)₆], as well as a control track (CTRL) [6 consecutive lysine AAG codons (AAG)₆] (Supplementary Fig.6A, see methods). Based on an AlphaFold rendered structure (version 2)^{37,38}, we chose residue 38 in an unstructured loop as an insertion site (Supplementary Fig.6B). Of note, (AAA)₆ and (AAG)₆ stretches are translated in the equal number of Lys residues, whereby (AAA)₆, but not (AAG)₆, stretches decrease ETFDH protein levels (Fig.6C). (AAG)₆ construct was thus used as a control to mitigate potential inadvertent effects of the insertion of poly-Lys on the ETFDH function. (AAA)₄ and (AAA)₆ stretches progressively reduced ETFDH protein abundance as compared to constructs harboring the (AAG)₆ track (Fig.6C). The reduction in ETFDH protein levels was mirrored by a gene-dose dependent increase in mTORC1 signaling, BCL-6 expression and reduction in 4E-BP1, but not 4E-BP2 protein levels (Fig.6C). Accordingly, *Eif4ebp1* mRNA levels were reduced in *ETFDH* dose-dependent manner, while *Eif4ebp2* mRNA levels remained unaffected (Fig.6D). Moreover, reduction in ETFDH protein levels correlated with increased mtDNA content (Fig.6E), oxygen consumption (Fig.6F), ATP production from OXPHOS (JATPox) (Fig.6G), bioenergetic capacity (Fig.6H), and cell proliferation (Fig.6I). Finally, ETFDH KO NT2197 tumors carrying (AAA)₆ insertions exhibit increased tumor growth and larger final tumor volumes relative to control tumors expressing (AAG)₆ insertions in SCID-BEIGE mice (Fig.6J-K). Altogether, these findings show that ETFDH gene-dosage influences neoplastic growth, thus suggesting that ETFDH may act as a haploinsufficient tumor suppressor.

Lastly, we determined whether the catalytic activity of ETFDH is required for its tumor suppressive activity. To achieve this, we re-expressed WT or ETFDH mutant (Y304A,G306E) with disrupted catalytic activity⁸ in HCT-116 ETFDH KO cells. Notwithstanding that both ETFDH variants were expressed to comparable levels, in contrast to WT, ETFDH (Y304A,G306E) mutant failed to suppress mTORC1 signaling and decrease 4E-BP1 levels (Supplementary Figure 6C), reduce proliferation (Supplementary Fig.6D) or perturb the bioenergetic profiles of HCT-116 ETFDH KO cells (Supplementary Fig.6E-F). Taken together, these data suggest the catalytic activity of ETFDH is required for its tumor suppressive actions.

Discussion

Notwithstanding that ETFDH mutations play a detrimental role in MADD metabolic disorders, we show that ETF insufficiency caused by reduced *ETFDH* expression plays a major role in rewiring cancer metabolism and signaling to stimulate neoplastic growth. Reduction of ETFDH levels appears to be a common feature of cancer cells and is likely a consequence of DNA hypermethylation (Fig.6B), an event that is frequently observed among tumor suppressor genes³⁹. Indeed, ETFDH was found among the most under-expressed metabolic enzymes in a study that compiled cancer microarray datasets⁴⁰. Moreover, recent studies highlighted crosstalk

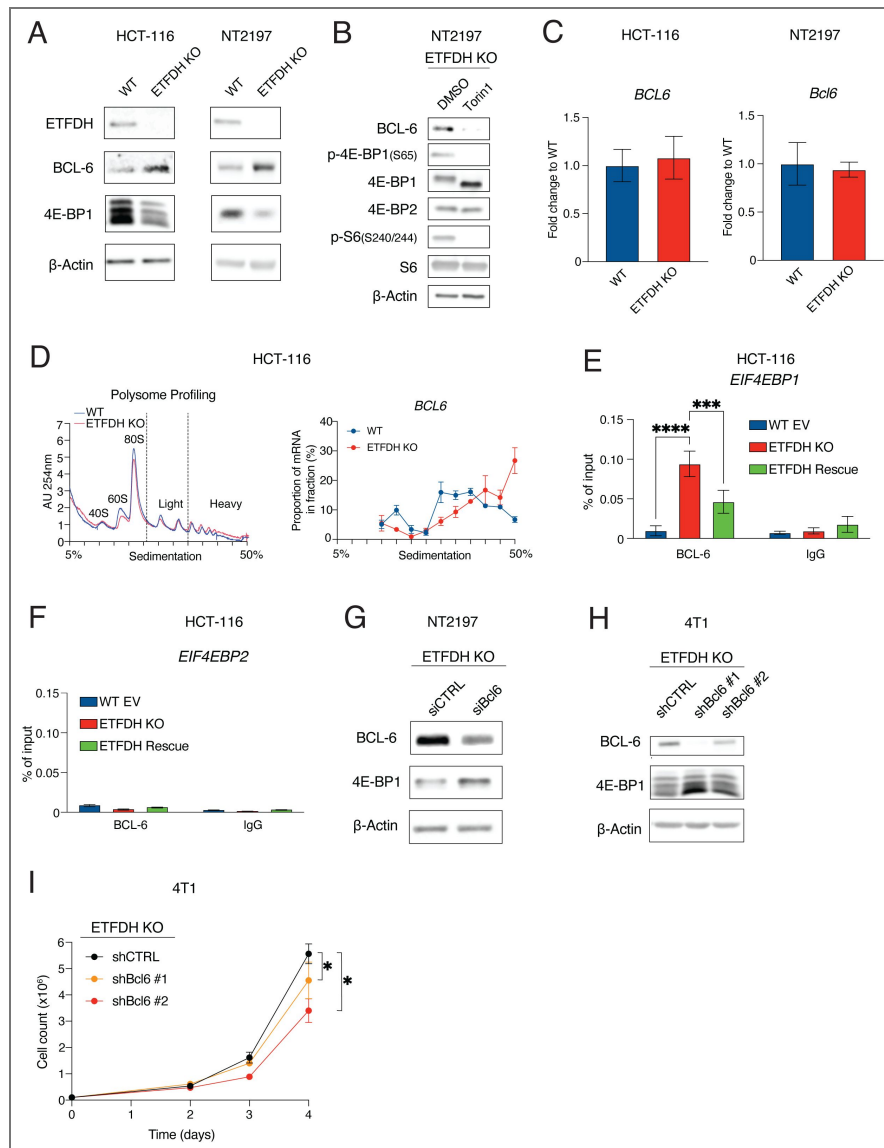


Figure 5. Loss of ETFDH increases BCL-6 levels leading to BCL-6-dependent inhibition of *EIF4EBP1* transcription.

A) Levels of indicated proteins in WT or ETFDH KO HCT-116 and NT2197 cells were determined by western blotting. β -Actin was used as a loading control (representative blots of n=3). B) Levels of indicated proteins in ETFDH KO NT2197 cells treated with DMSO or Torin1 (250nM) for 4 hours were determined by western blotting. β -Actin was used as a loading control (representative blots of n=3). C) *BCL6/Bcl6* mRNA abundance in WT and ETFDH KO HCT-116 and NT2197 cells was determined by RT-qPCR. *PP1A* mRNA was used as a control for HCT-116 experiments, while *Actb* mRNA was used as a control for NT2197 experiments. Data are presented as fold change in *EIF4EBP1/PP1A* and *EIF4EBP2/PP1A* (HCT-116) and *Eif4ebp1/Actb* and *Eif4ebp2/Actb* (NT2197) ratios relative to WT cells +/- SD (n=3), paired Student's *t* test. D) Polysome profiling of WT and ETFDH HCT-116 KO cells. The absorbance at 254nm (AU 254nm) was used to monitor distribution of the 40S-, 60S-ribosomal subunits, monosomes (80S) and polysomes across the gradient. *BCL6* mRNA abundance from polysome gradient fractions was obtained using RT-qPCR. Data are shown as a mean percentage of mRNA in each fraction relative to cumulative corresponding mRNA amount across the whole gradient +/-SD (n=3). E-F) Binding events of BCL-6 to the promoters of *EIF4EBP1* and *EIF4EBP2* were determined by ChIP-qPCR. IgG was used a negative control. Data are presented as % of input (n=3), *p<0.05, two-way ANOVA, Dunnett's post-hoc test. G) The levels of indicated proteins in ETFDH KO NT2197 cells transfected with siRNA targeting BCL-6 (siBCL-6) or control, scrambled siRNA (siCTRL) were determined by western blotting. β -Actin was used as loading control (representative blots of n=2). H) Levels of indicated proteins in ETFDH KO 4T1 cells infected with shRNAs targeting BCL-6 (shBCL-6 #1 or shBCL-6#2) or control, scrambled shRNA (shCTRL) were monitored by western blotting. β -Actin served as loading control (representative blots of n=3). I) Proliferation of ETFDH KO shCTRL, shBCL-6 #1, or shBCL-6#2 4T1 cells. Data are presented as cell count means (n=3), *p<0.05, one-way ANOVA, Dunnett's post-hoc test.

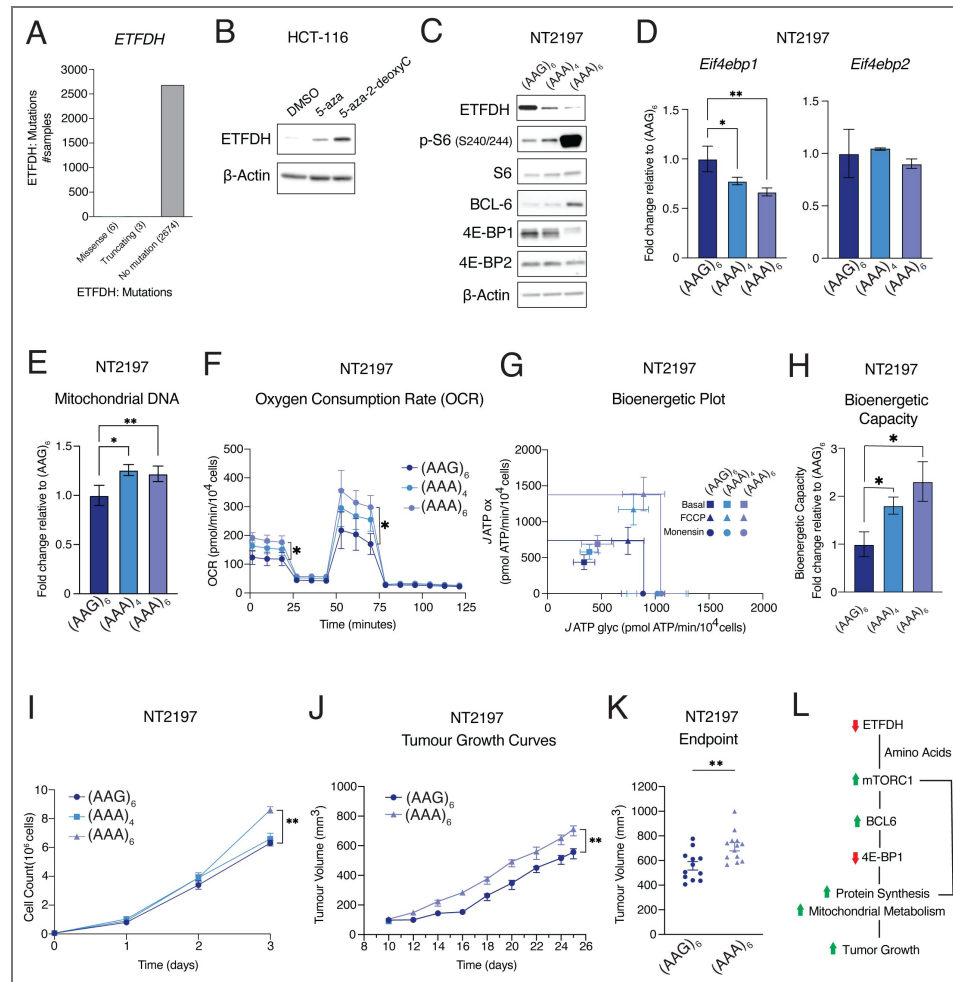


Figure 6. Reduced expression of ETFDH increases mitochondrial metabolism and tumor growth

A) ETFDH mutations from 2683 samples across multiple cancer types. Number of missense and truncating mutations are shown relative to samples with no mutations. Data are acquired from the ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium³⁴, accessed from the cBioPortal server (www.cbioportal.org). B) Levels of indicated proteins in WT HCT-116 cells treated with 2.5μM 5-azacytidine (5-aza), 10μM 5-aza-2'-deoxycytidine (5-aza-2-deoxyc), or a vehicle (DMSO) for 72 hours were determined by western blotting. β-Actin was used as a loading control (representative blots of n=3). C) Levels and the phosphorylation status of indicated proteins in ETFDH KO NT2197 cells expressing ETFDH harboring (AAG)₆, (AAA)₄, (AAA)₆ tracks were determined by western blotting. β-Actin was used as a loading control (representative blots of n=3). D) *Eif4ebp1* and *Eif4ebp2* mRNA abundance in (AAG)₆, (AAA)₄, and (AAA)₆ NT2197 cells were determined by RT-qPCR. *Actb* was used as a house-keeping gene. Data are presented as fold change in *Eif4ebp1/Actb* and *Eif4ebp2/Actb* ratios relative to (AAG)₆ NT2197 cells (n=3), *p<0.05, **p<0.01, one-way ANOVA, Dunnett's post-hoc test. E) Mitochondrial DNA content in (AAG)₆, (AAA)₄, and (AAA)₆ NT2197 cells were monitored by qPCR. Mitochondrial DNA (mtDNA) content was normalized to genomic DNA (gDNA) content. Data are presented as mean fold change relative to (AAG)₆ NT2197 cells ± SD (n=3), *p<0.05, **p<0.01, one-way ANOVA, Dunnett's post-hoc test. F) Oxygen consumption of (AAG)₆, (AAA)₄, and (AAA)₆ NT2197 cells was determined using Seahorse bioanalyzer. Data are normalized to cell count and presented as means ± SD (n=4), one-way ANOVA, Dunnett's post-hoc test. G-H) Bioenergetic Plot for Basal, FCCP, and Monensin / ATP fluxes (G) and Bioenergetic Capacity (H) derived from (AAG)₆, (AAA)₄, and (AAA)₆ NT2197 cells. Data are presented as means ± SD (n=4), one-way ANOVA, Dunnett's post-hoc test. I) Proliferation of (AAG)₆, (AAA)₄, and (AAA)₆ NT2197 cells. ETFDH rescue measurements are the same as in Supplementary Fig.1F. Data are presented as cell count means ± SD (n=4), **p<0.01, one-way ANOVA, Dunnett's post-hoc test. J-K) Tumor growth of (AAG)₆ and (AAA)₆ NT2197 cells following mammary fat-pad injection (J) and endpoint tumor volumes (K). Growth was assessed using calipers. Data are presented as means ± SEM ((AAG)₆ n=6-12, (AAA)₆ n=11-13), **p<0.01, unpaired Student's t test. L) Schematic representation of the model whereby reduction in ETFDH levels accelerates tumor growth. Reduction of ETFDH triggers accumulation of intracellular amino acids thus increasing mTORC1 signaling. This is orchestrated with BCL-6-dependent reduction in 4E-BP1 levels to drive mitochondrial biogenesis and increase bioenergetic capacity of cancer cells ultimately leading to more aggressive tumor growth.

between aberrant mTOR signaling and epigenetic perturbations^{41–44}, which further reinforces the plausibility that epigenetic reprogramming in cancer cells may underpin the pro-neoplastic role of ETFDH downregulation and/or explain resulting phenotypes.

Intriguingly, although ETF insufficiency limits the flexibility of cancer cells in using fuels (e.g., lipids and amino acids) for OXPHOS, it triggers signalling events that establish a feed-forward loop via mTORC1, leading to increased BCL-6 levels and reduced *EIF4EBP1* transcription (Fig. 6L [↗](#)). ETF insufficiency thus orchestrates enhanced mTORC1 signaling with downregulation of 4E-BP1 protein levels to reprogram protein synthesis and ultimately increase mitochondrial biogenesis and functions thereby providing a selective proliferative advantage to cancer cells (Fig. 6L [↗](#)). To this end, ETF insufficiency allows cancer cells to trade their metabolic flexibility for increased bioenergetic capacity and a re-wired signaling state that favors neoplastic growth. Importantly, while ETFDH plays an essential role in muscle, we did not observe a major effect of ETFDH disruption on proliferation, mTOR signaling or 4E-BP1 levels in non-transformed murine breast epithelial NMuMG cells. These observations suggest that ETF insufficiency causes remodeling of signaling and metabolic networks that are favourable to cancer cells.

Our findings are consistent with previously attributed tumor-suppressive effects of 4E-BP1^{45–50}. More recently, it has been demonstrated that under certain contexts 4E-BPs may play a beneficial role for tumor survival. For instance, 4E-BPs may facilitate tumor growth by regulating global and selective mRNA translation under periods of cellular stress^{51,52}, including suppression of fatty acid biosynthesis under glucose deprivation^{53,54}. Collectively, these findings demonstrate that 4E-BPs are likely to play more complex, context dependent roles during cancer progression than previously anticipated. Moreover, the role of different 4E-BP proteoforms in cancer was in general thought to be redundant. Our findings however demonstrate that in the context of ETF insufficiency selective decrease in 4E-BP1, but not 4E-BP2 levels plays a role in driving neoplastic growth thereby suggesting that 4E-BP1 and 4E-BP2 may play hitherto unappreciated non-overlapping roles in neoplasia.

Several metabolites can be sensed via signaling partners upstream of mTORC1, including leucine, arginine, methionine/SAM, and threonine⁵⁵. Branched-chain amino acids (leucine, isoleucine, and valine), which are among the highest upregulated metabolites in ETFDH deficient cells (Fig. 3A [↗](#)) serve as ETFDH substrates, and have been described to display strong activation capabilities towards mTORC1 in the literature^{56,57}. Glutamine can also activate mTORC1 through Arf family of GTPases⁵⁸. Indeed, glutamine can supplement the non-essential amino acid (NEAA) pool through transamination⁵⁹ and amino acid uptake⁶⁰. Accordingly, the maintenance of NEAA that are non-ETFDH substrates may be supported by the global metabolic rewiring fueled by enhanced glutamine metabolism in ETFDH-deficient cells. Deciphering the mechanisms leading to accumulation of specific amino acids and their role in ETFDH-dependent mTORC1 modulation is warranted.

In conclusion, we show that ETF insufficiency caused by reduced ETFDH expression remodels signaling and mitochondrial metabolism in a manner that drives neoplastic growth (Fig. 6L [↗](#)), while either being neutral (NMuMG) or deleterious (muscle cells) in non-malignant cells. These findings explain how disruption of mitochondrial enzyme, rather than reducing cellular fitness, is common in cancer because it increases mitochondrial energy flux to drive neoplastic growth.

Methods

Cell lines and cell culture

HCT-116, NALM6, and HEK293T cells were obtained from American Type Culture Collection (ATCC) and verified using ATCC Cell line authentication service. NMuMG, NT2197, and MT4788 Parental and STAT1 KO cells were obtained by Dr. J. Ursini-Siegel's lab, while 4T1 (4T1-526) cells were obtained from Dr. P.M. Siegel's group. HEK293T and HCT-116 cells were cultured in DMEM supplemented with 10% fetal bovine serum (FBS), 1% penicillin/streptomycin and supplemented with 2mM L-Glutamine to obtain a final concentration of 6mM L-Glutamine. NALM6 and 4T1 cells were cultured in RPMI-1640 supplemented with 10% FBS, 1% penicillin/streptomycin and 2mM L-

Glutamine, NMuMG and NT2197 cells were cultured in DMEM supplemented with 10% FBS, 1% penicillin/streptomycin, 2mM L-Glutamine, 10 mg/mL insulin, 20 mM HEPES, pH 7.5. NT2197 culture media was supplemented with puromycin (2 μ g/mL) as previously described¹¹. MT4788 were grown in DMEM, 2.5% FBS, mammary epithelial growth supplement (MEGS), 1% penicillin/streptomycin and gentamycin as previously described²⁵. All cell lines were maintained at 37°C and 5% CO₂ in a humidified environment and tested periodically for mycoplasma according to the manufacturer's instructions.

Animals

6–8-week-old SCID-BEIGE mice were purchased from Charles River. Male SCID-BEIGE mice were used for intracaecal experiments, while female mice were used for mammary fat pad injections. All animal experiments were performed in accordance with the guidelines of the McGill University Animal Ethics Committee and the Canadian Council on Animal Care as approved by the facility animal care committee.

Xenograft Experiments

Luciferase-expressing HCT-116 cells (250,000 cells) were injected intracaecally in male (8–10 weeks old) SCID-BEIGE mice as previously described⁶¹. Tumor growth was monitored and assessed by luminescence (IVIS spectrum, PerkinElmer). Animals were sacrificed after 50 days. Quantification of signal intensity was performed with Living Image software (PerkinElmer). NT2197 mammary tumor cells (50,000 cells) were injected in mammary fat pads of 6–8-week female SCID-BEIGE mice. Tumor growth was monitored using calipers. Measurements were shown as once tumor volume reached 100–200 mm³ and animals were sacrificed before tumors reached 1cm³. Tumors were excised, placed in 10% formalin, and embedded for immunohistochemistry.

Cell Proliferation Assay

Cells were counted using a Countess automated cell counter (Invitrogen) and trypan blue exclusion was used for determining cell viability. Cells (HCT-116, NT2197, NMuMG, 4T1, NALM-6) were seeded in 6-well dishes and counted to 3–5 days. HCT-116 cells were seeded on day 0 at a density of 100,000 cells for Supplementary Figs.1E [1E](#), 3C, 3E-G [3E-G](#), 4D [4D](#) and 50,000 cells for Supplementary Fig.6D [6D](#). NT2197 cells were seeded on day 0 at a density of 60,000 cells in Fig.6I [6I](#) and Supplementary Fig.1F. NMuMG cells were seeded on day 0 at a density of 60,000 cells in Supplementary Fig.1I. 4T1 cells were seeded on day 0 at a density of 60,000 cells in Fig.5I [5I](#) and Supplementary Fig.1G. NALM-6 cells were seeded on day 0 at a density of 20,000 cells in Supplementary Fig.1H. For glutamine deprivation experiments, HCT-116 cells were seeded in full culture for 24 hours. The next day, cells were grown in media in the presence or absence of glutamine for 48 hours and then counted. For Torin1/Bis-35x treatment experiments, HCT-116 cells were seeded overnight. The next day, media was replaced with treatment media containing Torin1 (50nM, 100nM, 250nM, 500nM) or Bis-35X (1nM, 10nM, 100nM, 1000nM) or DMSO for 72 hours and then counted.

Immunohistochemistry

Immunohistochemical staining of colon human paraffin-embedded tissue array (TissueArray.com [1](#), CO992b) was performed as previously described⁶². Briefly, the tissue slide was baked for 55°C for 30 min followed by incubation in xylene (2× 5 min), 100% ethanol (2× 5 min), 95% ethanol (2× 5 min), 75% ethanol (1× 5min), and ddH₂O (2× 5min). The slide was submerged in 10mM sodium citrate buffer (pH 6.0), 0.05% Tween-20 and heat-induced epitope retrieval was carried out in a pressure cooker for 30 min. The slide was cooled for 20 min at room temperature, rinsed in ddH₂O for 5 min, and washed with 1xTBS + 0.3% Triton X-100 for 10 min, then 1xTBS for 5 min at room temperature. The slide was incubated in 3% H₂O₂ for 10 min at room temperature and washed with TBST (1xTBS + 0.1% Tween-20) for 5 min to inactivate peroxidases. A liquid blocker pen (PAP Pen, DAKO) was used to circle tissues on the slide. The TMA was blocked for 1 hour in room temperature in the presence of DAKO Protein Block-Serum Free

(DAKO, Cat#X0909). The slide was incubated overnight at 4°C with the ETFDH antibody (1:500, ProteinTech 11109-1-AP). The next day, the slide was washed in TBST (3x 5min), and secondary incubation was performed for 30 min at room temperature with secondary antibody (Antibody background reducing diluent and HRP rabbit substrate (Cell Signaling, Cat#7074)). The slide was washed in TBST (3x 5min) and DAB (Di-amine-benzidine) peroxidase substrate (Vector, Cat# SK-4100) was applied to the slide for 5 min. Reaction was stopped by washing in ddH₂O for 5 min. The slide was counterstained with filtered hematoxylin for 20 sec and rinsed with ddH₂O for 5 min. The slide was placed in 75% ethanol (1 min), 95% ethanol (1 min), 100% ethanol (1 min) and xylene (2x 5min). The slide was mounted with a cytooseal at 60°C and dried in a fumehood overnight. Quantitation of sections was performed using Imagescope software (Aperio).

Western Blotting

Western blotting was performed as previously described⁴⁴. Briefly, cells were washed twice with PBS and lysed with RIPA buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.5% sodium deoxycholate, 1 mM PMSF, 1 mM DTT, 1% NP40, 0.1% SDS), supplemented with 1X protease inhibitor (Roche) and 1x PhoSTOP (Roche). Lysates were cleared at 4°C (10min, 13,000rpm). Protein concentrations were determined using BCA kit (ThermoFisher). Lysates were mixed with 5x Laemmli buffer (60mM Tris-HCl (pH 6.8), 10% glycerol, 2% SDS, 5% 2-mercaptoethanol, 0.05% bromophenol blue) and boiled (95°C for 5min). Proteins were resolved by SDS-PAGE (Bio-Rad) and transferred via wet transfer apparatus (Bio-Rad) to nitrocellulose membranes (Amersham). Membranes were blocked in 5% BSA (dissolved in TBST (0.1% Tween 20 in 1× TBS)) and incubated in primary antibodies overnight at 4°C. The next day, membranes were washed in TBST and incubated with HRP-conjugated secondary antibodies for 1 hour. Membranes were washed and applied with ECL (Bio-Rad) for 1min. Exposure was carried out on an Azure c300 (Azure Biosystems). Images were quantified by ImageJ and analysed using GraphPad Prism10. The list of primary antibodies is described in Supplementary Table 1 [1](#).

For λ -phosphatase experiments, cells were lysed with RIPA and supplemented with 1X protease inhibitor, without 1x PhoSTOP. Protein samples were combined with NEBuffer for protein metallophosphatases (PMP), MnCl₂, and λ -protein phosphatase for 30min at 30°C according to the manufacturer's instructions (New England Biolabs).

Cap-Binding Pull-Down Assay

Assay was carried out as previously described¹⁸. Briefly, NT2197 cells were seeded overnight in 150mm plates. The next day, media was changed, and cells were washed with PBS and dissolved in lysis buffer containing 50mM MOPS/KOH (7.4), 100mM NaCl, 50 mM NaF, 2mM EDTA, 2mM EGTA, 1% NP40, 1% sodium deoxycholate, 7 mM β -mercaptoethanol, supplemented with 1X protease inhibitor (Roche) and 1x PhoSTOP (Roche). Lysates were cleared at 4°C (10min, 16,000g). Protein concentrations in samples were elucidated by BCA (ThermoFisher). Samples were equilibrated for 20 min rotating at 4°C with m⁷-GDP-agarose beads (Jena Bioscience), then washed in buffer containing 50mM MOPS/KOH (7.4), 100mM NaCl, 50 mM NaF, 0.5mM EDTA, 0.5 mM EGTA, 7 mM β -mercaptoethanol, 0.5 mM PMSF, 1mM Na₃VO₄ and 0.1mM GTP, by centrifugation (500g for 1min). Bound proteins were eluted from beads via boiling in loading buffer. Western blotting was performed on eluted proteins and input samples to assess for eIF4F complex formation using 4E-BP1, eIF4G1, eIF4E, and β -actin antibodies.

Flow Cytometry

NT2197 cells were seeded in 6-well dishes for 24 hours. The next day, cells were stained with 50nM MitoTracker Deep Red (ThermoFisher) for 15 minutes at 37°C in the dark. Cells were washed, counted, and 100,000 cells were dissolved in 500 μ l HBSS (Gibco). Samples were analyzed with the LSR Fortessa cytometer (Becton Dickinson, Mountain View, CA). Fluorescence intensity was detected by excitation at 644nm and acquisition on the 665/-A channel for MitoTracker Deep Red. ROS measurements were performed as previously described¹⁶². Briefly, cells were incubated with

H₂DCFDA (Molecular Probes) for 30 minutes at 37°C. Cells were trypsinized, washed, and resuspended in 500µl HBSS. Samples were sorted on a BD FACS Canto II system. Fluorescence intensities were calculated by FlowJo (Tree Star, Inc.).

Mitochondrial DNA Quantitation

Quantitation of mitochondrial DNA was carried out as previously described²¹. HCT-116 or NT2197 cells were seeded in 6-well dishes for 24 hours. Genomic and mitochondrial DNA were extracted using PureLink Genomic DNA Mini Kit (ThermoFisher) and quantified by qPCR using SensiFAST™ SYBR® Lo-ROX kit (Bioline).

Polysome-Profiling Assay

Experiments were carried out as previously described⁶³. Briefly, HCT-116 cells were seeded overnight in 150mm plates. The next day, cells were washed in PBS containing 100 µg/ml cycloheximide and lysed in hypotonic buffer (5 mM Tris-HCl [pH 7.5], 2.5 mM MgCl₂, 1.5 mM KCl, 100 µg/ml cycloheximide, 2 mM DTT, 0.5% Triton X-100, and 0.5% sodium deoxycholate). Samples were loaded onto 10%-50% wt/vol sucrose density gradients (20 mM HEPES-KOH pH 7.6, 100 mM KCl, 5 mM MgCl₂) and spun on a SW40Ti rotor (Beckman Coulter) at 4°C (36,000 rpm for 2 hours). Samples were fractionated and recorded at OD 254nm using an ISCO fractionator (Teledyne ISCO). RNA from fractions and input was extracted using TRIzol (ThermoFisher) according to the manufacturer's instructions. RNA from each fraction and input was isolated using TRIzol (Invitrogen) according to the manufacturer's instructions. RT-qPCR was performed on fractions and input RNA (refer to section "RNA extraction and RT-qPCR" for more information). Primers are listed in [Supplementary Table 2](#).

Puromycilation Assay

HCT-116 cells were grown in 10cm dishes for 24 hours. The next day, cells were treated with puromycin (10ug/ml) for 20 minutes. Cells treated with DMSO were used as negative controls. Cells were lysed and protein samples were quantified. Western blotting was performed (refer to "Western Blotting" section for more information) and membranes were incubated in anti-puromycin antibody (Millipore, Cat#MABE343). Quantification of bands was performed by ImageJ.

Stable Isotope Tracing

Metabolite detection by gas chromatography-mass spectrometry (GC-MS) and stable isotope tracing analysis (SITA) was performed as previously described⁶⁴. Briefly, NT2197 cells were seeded in 6-well dishes. Prior to ¹³C-glutamine SITA experiments, culture media was replaced with one whose composition contains 6mM unlabelled glutamine for 2 hours. Afterwards, media was replaced with media containing 6mM labelled ([U-¹³C])-glutamine media (Cambridge Isotope Laboratories, MA, USA; CLM-1822; L-glutamine ([U-¹³C5]), 99%) for 5, 15, 30, and 60 minutes. For ¹³C-leucine SITA experiments, culture media was replaced with one containing unlabelled leucine (0.105g/L) for 2 hours, then incubated in media containing labelled (0.105g/L) ([U-¹³C])-leucine media (Cambridge Isotope Laboratories, MA, USA; CLM-2262-H-0.1; L-leucine ([U-¹³C6]), 99%) for 24 hours. Both steady state and tracing samples were washed in cold saline solution (9g/L NaCl) and scraped off with 80% methanol. Samples were sonicated at 4°C for 10 minutes (high setting, 30 seconds on/30 seconds off cycles) to rupture cells. Lysates were centrifuged at 4°C (14000g, 10 minutes), and supernatants were collected with addition of an internal standard (750ng myristic acid-D27). Samples were dried overnight at 4°C by speed-vac (Labconco). Dried pellets are resuspended in methoxyamine hydrochloride (10mg/ml), sonicated, and cleared for 10 minutes. Samples are heated 70°C for 30 minutes then applied with N-tert-butyldimethylsilyl-N-methyltrifluoroacetamide (MTBSTFA) at 70°C for 1 hour. Derivatized samples are injected into an Agilent 5975C GC/MS (Agilent Technologies, CA, USA) using methods described previously⁶⁴. ¹³C-tracer samples were run in parallel with unlabelled samples. MassHunter software (Agilent) was

used to carry out mass isotopomer distribution analyses, with metabolites normalized to myristic acid-D27 and cell number. Isotopic distributions were corrected for naturally occurring isotopes using an in-house algorithm⁶⁵.

Steady-State Analysis of Nucleotides

Steady-state nucleotide abundances were determined using liquid chromatography-mass spectrometry (LC-MS/MS). Cells were washed with 150 mM ammonium formate solution (4°C, pH 7.4), scraped, and extracted with 600 µl of methanol/acetonitrile solution (31.6% MeOH/36.3% ACN). Cell lysis and homogenization was conducted by bead-beating with 4 ceramic beads (2 mm) for 30 seconds at 50 Hz using a TissueLyser II (Qiagen). Dichloromethane was added to samples, which were subsequently centrifuged to separate extracts into aqueous and organic layers. Supernatants from aqueous layers were dried by vacuum centrifugation at -4°C overnight (Labconco). Dried samples were re-suspended in 50 µl of water and centrifuged at 1°C. 5 µl of sample was injected onto an Agilent 6470 Triple Quadrupole (QQQ)-LC-MS/MS (Agilent Technologies). Separation of metabolites was conducted using a 1290 Infinity ultra-performance quaternary pump liquid chromatography system, with a mass spectrometer equipped with a Jet StreamTM electrospray ionization source operating in negative mode (Agilent Technologies), with the following settings: source-gas temperature (150 °C), flow (13 L min⁻¹), nebulizer pressure (45 psi), and capillary voltage (2,000 V). Chromatographic resolution of metabolites was conducted using a Zorbax Extend C18 column 1.8 µm, 2.1 × 150mm² with guard column 1.8 µm, 2.1 × 5mm² (Agilent Technologies). The gradient commenced at 100% mobile phase A (97% water, 3% methanol, 10 mM tributylamine, 15 mM acetic acid, 5 µM medronic acid) for 2.5 min, followed with a 5-min gradient to 20% mobile phase C (methanol, 10 mM tributylamine, 15 mM acetic acid, 5 µM medronic acid), a 5.5-min gradient to 45% C and a 7-min gradient to 99% C at a flow rate of 0.25 mL min⁻¹. This was followed by a 4-min hold time at 100% mobile phase C. The column was restored by washing with 99% mobile phase D (90% ACN) for 3 min at 0.25 mL min⁻¹, followed by increase of the flow rate to 0.8 mL min⁻¹ over 0.5 min and a 3.85-min hold, after which the flow rate was decreased to 0.6 mL min⁻¹ over 0.15 min. The column was subsequently re-equilibrated at 100% A over 0.75 min, during which the flow rate was decreased to 0.4 mL min⁻¹, and held for 7.65 min. The flow was brought back to forward flow at 0.25 mL min⁻¹ one minute before the next injection. The column temperature was maintained at 35°C. Data were analyzed by using MassHunter Quantitative Data Analysis B.10.00 (Agilent Technologies). Data presented are peak area normalized by cell number. Authentic metabolite standards (Sigma Aldrich) were used to obtain reaction monitoring parameters (qualifier/quantifier ions and retention times). All LC/MS grade solvents and salts are purchased from Fisher.

RNA Extraction and RT-qPCR

HCT-116 and NT2197 cells were plated in 6-well dishes and extracted using TRIzol (Ambion) or Aurum Total RNA Mini Kit (Bio-Rad) following the manufacturers' instructions. cDNA was synthesized from purified RNA using SensiFAST cDNA Synthesis Kit (Bioline), as per manufacturer's protocol. RT-qPCR was performed using SensiFAST SYBR Lo-ROX kit (Bioline). Primer sequences are found in Supplementary Table 2 [2](#).

Chromatin Immunoprecipitation (ChIP)

Experiments were performed as previously described⁴⁴. Briefly, HCT-116 cells were grown in 150mm dishes and fixed in 4% formaldehyde for 10 min, followed by centrifugation. Pellets were resuspended in ChIP-buffer (50 nM Tris (pH 8), 100 mM NaCl, 5 mM EDTA, 1X PMSF, 2 mM NaF, 0.25% NP-40, 0.25% Triton X-100, 0.25% Sodium Deoxycholate, 0.005% SDS) supplemented with 1x cOmplete protease inhibitor (Roche). Samples are sonicated using Sonic Dismembrator Model 500 (ThermoFisher) at 5 cycles at 20% power, 5 cycles at 25% power, 5 cycles at 30% power; each cycle is 10 sec. Lysates are spun at 4°C (14,000rpm, 10min) and protein concentration was measured from supernatants using the BCA kit (ThermoFisher). Samples were loaded onto Protein G Plus-Agarose Suspension Beads (Millipore) and pre-cleared at 4°C for 3 hours. Input samples were

collected, and immunoprecipitation was performed with anti-BCL6 (Cell Signaling, #5650) or anti-IgG (Cell Signaling, #2729) on the remainder of the sample overnight at 4°C. The following day, samples were washed in three wash buffers (20 mM Tris (pH 8), 2 mM EDTA, 0.10% SDS, 1% Triton X-100 with 150/200/599mM for Wash1/2/3 buffers respectively), then a wash with LiCl buffer (10 mM Tris (pH 8), 1 mM EDTA, 0.25 M LiCl, 1% NP-40, 1% Sodium Deoxycholate), and two washes with TE buffer (10 mM Tris (pH 8), 1 mM EDTA). Samples were eluted in elution buffer (0.1 M NaHCO₃, 1% SDS), and de-crosslinked at 65°C overnight. The following day, proteinase K was applied to samples and heated at 42°C for 1 hour. DNA was purified and collected using DNA collection column (BioBasic). ChIP-qPCR was performed for *EIF4EBP1* and *EIF4EBP2* using sequences found in [Supplementary Table 3](#).

Respirometry Assay

Oxygen consumption (OCR) and extracellular acidification rate (ECAR) were measured using Seahorse XFe24 and XFe96 analyzers (Agilent) with the Mito Stress Test (Agilent) and Palmitate Oxidation Assay (Agilent), according to the manufacturer's instructions. Briefly, HCT-116 and NT2197 cells were seeded in Seahorse culture plates (XFe24: 50,000 cells or XFe96: 20,000 cells) and incubated at 37°C. Cells were washed twice and incubated in assay media. For mito stress test experiments, assay media is composed of 10mM glucose, 2mM glutamine, and 1mM sodium pyruvate. Injections include oligomycin (1μM), FCCP (1μM), rotenone/antimycin A (1μM), and monensin (20μM). ATP production from OXPHOS (ATPox) or glycolysis (ATPglyc) were quantified using algorithms presented previously⁶⁶. For palmitate oxidation assay, cells were seeded in substrate-limited media (DMEM, 1% FBS, 0.5mM glucose, 1mM glutamine, 0.5mM carnitine). The next day, cells are washed with FAO Buffer (includes 0.5mM glucose, 0.5mM carnitine, and 5mM HEPES). Immediately prior to run, palmitate-BSA or BSA control is added to the cell media. Basal respiration was recorded and values were normalized to cell counts.

Soft Agar/Colony Formation Assay

Experiments were conducted as previously shown¹². Briefly, 6-well dishes are filled with a bottom layer solution (1:1; 1% noble agar and 2X culture media (2X DMEM, 20% FBS, 2% Pen/Strep/Glutamine)) per well. Dishes are incubated at room temperature for 30 min to allow solution to solidify. 5,000-10,000 HCT-116 and NT2197 cells are mixed 1:1 with 0.6% noble agar and 2X culture media. 1.5ml of this mixture is transferred on top of the solidified agar wells. Dishes are incubated at room temperature to allow the top layer to solidify. Cells are grown at 37°C until colony formation is present. Dishes were stained with nitroblue tetrazolium (Sigma), and colonies were counted.

Protein Stability Assay

HCT-116 cells were seeded in 10cm dishes. The next day, cells were treated with cycloheximide (CHX) (50μg/ml) or DMSO for 2 hours. Furthermore, cells were treated with the proteasomal inhibitor, MG-132 (10μM) or DMSO for 2 hours. Final time point for all conditions was 4 hours. Protein was extracted, quantified, and western blotting was performed for ETFDH, p21, 4E-BP1, or β-actin (refer to “Western Blotting” section for more information).

RNA Stability Assay

Assay was performed as previously described⁶⁷. Briefly, HCT-116 cells were seeded in 6-well dishes. The next day, cells were treated with actinomycinD (ActD) for 2,4,8 and 24 hours. RNA was extracted, cDNA was synthesized, and RT-qPCR was performed (refer to “RNA extraction and RT-qPCR” for more information).

Generation of ETFDH Knockout Cell Lines

To deplete endogenous ETFDH expression in HCT-116 and NALM6 cells, two guides RNA (gRNAs) targeting exon 6 and exon 9 of human ETFDH (against ETFDH sequences AGGTTGGCCGAATGCTAG GATGG and GATGTAGGGATACAAAAGGATGG) (refer to [Supplementary Table 4](#) for gRNA

sequences) were designed using CHOPCHOP (<https://chopchop.cbu.uib.no/>)⁶⁸ and purchased with the appropriate overhangs to be cloned into LentiCRISPRv2(GFP) (Addgene #82416). Cloning was carried out as previously described^{69,70}, with the following modifications: BsmBI-v2 (New England Biolabs) was used to digest the lentiCRISPRv2 plasmid; plasmid DNA was recovered using Zymoclean Gel DNA Recovery Kit (Zymo Research), and ligation was performed overnight at 4°C. Transformation of the ligated plasmid was carried out at 42°C in chemically competent Stbl3 cells (Fisher Scientific). Bacteria were plated on Lysogeny Broth (LB)-agar plates containing ampicillin (1 µg/mL) (BioBasic). Extraction of plasmid DNA was conducted using the QIAprep Spin Miniprep Kit (Qiagen) and samples were sequenced using primer for human U6 promoter (refer to Supplementary Table 5 [for sequence](#)). Nucleofection was used to deliver the validated lentiCRISPRv2 plasmids into HCT-116 and NALM6 cells. Transfections of the plasmids were carried out in Nucleofector cuvettes, according to the optimized protocols for NALM6 and HCT-116 cells (Lonza Bioscience), with 1×10^6 cells used and 1X PBS was used instead of the commercial 4-D Nucleofector solution. Post-nucleofection, cells were allowed to grow for 72 hours at 37°C, 5% CO₂, prior to cell sorting (BD FACSaria™ Fusion Flow Cytometer (BD Biosciences)). Cells expressing GFP were selected and grown in 96-well plates containing conditioned media (45% filtered culture media, 45% fresh media, & 10% FBS). After two weeks, sorted clones were transferred into 24-well plates and later into 6-well plates to be collected and validated. Depletion of endogenous ETV6 protein levels in both HCT-116 and NALM-6 cells was initially confirmed by immunoblotting. Candidate clones with loss of ETV6 expression were kept for further validation. Moreover, the genomic DNA loci of candidate clones were sequenced and analyzed for insertion–deletion mutations (INDELs). Two sets of PCR primers were designed in regions flanking the CRISPR target sites (refer to Supplementary Table 5 [for validation PCR sequences](#)). Extraction of genomic DNA for each clone was carried out using the PureLink Genomic DNA Mini Kit (Fisher Scientific). Two rounds of polymerase chain reaction (PCR) on the extracted genomic DNA (35 PCR cycles with Set 1 followed by 45 PCR cycles with Set 2) were carried out and gel electrophoresis was utilized to separate the PCR products. The smaller PCR product for each clone was excised and purified using the Zymoclean Gel DNA Recovery Kit (Zymo Research). The products were sequenced, and chromatograms were analyzed for the presence of mutations causing a premature stop-codon or a frameshift. Insertion–deletion mutations (INDELs) in *ETV6* were verified by PCR followed by sequencing, while the absence of ETV6 protein was confirmed by western blotting for HCT-116 cells and NALM6 cells.

To deplete ETV6 expression in NMuMG, NT2197 and 4T1 cells, two guides (gRNAs) targeting exon 2 of mouse *Etv6* (against mEtv6 sequences 5'-ATTTTATGCGAGCTACTGG-3' and 5'-GAACATCTGGAGCACAGAGG-3') were designed using CHOPCHOP and cloned in LentiCRISPRv2(GFP) (Addgene #82416) and sequenced as shown above. For lentiviral production, 2.5×10^5 HEK293T cells were seeded overnight in a 6-cm dish (Sarstedt). The next day, cells were co-transfected with 4µg of respective lentiCRISPRV2 GFP plasmid containing either one of the gRNA inserts or empty-lentiCRISPRV2 GFP plasmid, 2.66µg of psPAX2 packaging plasmid (Addgene 35002), and 1.66 µg of pMD2.G plasmid (Addgene #12259) using the jetPRIME transfection reagent as described by the manufacturer's protocol (Polyplus transfection). Growth media was changed after 24 hours. After 48 hours, viral supernatant was filtered (0.45 µm filter; (Frogga Bio)), mixed 1:1 with fresh culture media, and applied to target cells grown in 6-well plates along with 8 µg/mL polybrene (Sigma-Aldrich). Cells were re-transduced the following two days with viral supernatant. Cells were allowed to recover for 48 hours post-transduction. Cells were sorted into 96-well plates (Sarstedt) and grown to generate stable cell lines, as shown above. Depletion of mouse *Etv6* protein levels was initially confirmed via immunoblotting for NT2197, 4T1, and NMuMG. The presence of INDELs leading to a premature stop codon or frameshift was determined by sequencing as shown above. Two sets of PCR primers were designed in the regions flanking the CRISPR target sites (refer to Supplementary Table 5 [for validation PCR sequences](#)).

Generation of ETFDH Rescue Cell Lines

To rescue human ETFDH expression in HCT-116 cells, ETFDH(NM_004453) Human Tagged ORF was cloned into pLenti-ETFDH-C-Myc-DDK-P2A-Puro plasmid according to the manufacturer's instructions (Origene) to generate the resulting plasmid, pLenti-ETFDH-C-Myc-DDK-P2A-Puro. To re-express mutant ETFDH (Y304A, G306E), WT ETFDH, or ORF-Stuffer lentiviral plasmids conferring hygromycin B resistance were designed using VectorBuilder (VB250613-1532mdm, VB250613-1530fzr, and VB900172-2887sfj respectively). Plasmids were transfected with psPAX2, pMD2.G into HEK293T cells with JetPrime reagent (Polyplus) according to the manufacturer's instructions. Viral supernatants were filtered (0.45µM) and applied to HCT-116 ETFDH KO cells with polybrene (6µg/mL). After 6 hours, cells were re-infected overnight. After 2 days, selection with puromycin (4µg/ml) was performed for 72 hours. Uninfected cells were selected with puromycin to serve as negative controls. The re-introduction of ETFDH expression or control plasmid in the knockout cells was confirmed via western blotting. To rescue ETFDH expression in NT2197 ETFDH KO cells, a lentiviral plasmid conferring hygromycin B resistance and expressing mEtfdh with PAM site mutations or expressing an ORF-Stuffer was created using VectorBuilder (VB221206-1257xet and VB900123-2599cba respectively). Lentivirus production and transduction were carried out as previously mentioned for three consecutive days. After 48h of recovery, cells were selected with 1 mg/mL hygromycin B for 3 days. Cells were confirmed by western blotting.

Generation of Poly(A) Track ETFDH Variants

Mutants were designed by inserting poly(A) tracks of multiple adenosines: control track (CTRL) [6 consecutive lysine AAG codons (AAG)₆], 12 adenosines [12A, equivalent to 4 consecutive AAA (Lys) codons (AAA)₄]; 18 adenosines [18A, equivalent to 6 consecutive AAA (Lys) codons (AAA)₆] within the rescue plasmid pLV[Exp]-Hygro-EF1A>mEtfdh. The polyA track insertion site was determined using AlphaFold (version 2)^{37,38} to minimize unintended effects caused by the poly-lysine stretches on the possible mitochondrial import sequence. The polyA tracks were inserted following residue 38, in an unstructured loop of the Etfdh protein. Site-directed mutagenesis was performed using QuikChange II site-directed mutagenesis kit, according to the manufacturer's instructions (Agilent). Briefly, cycling parameters are followed as: Cycle 1: 95°C for 1min; Cycle 2: 95°C for 30 sec, 60°C for 30 sec, 68°C for 10min; Cycle 3: 68°C for 7min. The resulting plasmids were generated: pLV[Exp]-Hygro-EF1A>mEtfdh-AAG₆, pLV[Exp]-Hygro-EF1A>mEtfdh-AAA₄, pLV[Exp]-Hygro-EF1A>mEtfdh-AAA₆. To generate the poly(A) track Etfdh mutant lines in NT2197 cells (termed AAG₆, AAA₄, AAA₆), lentivirus production and transduction into NT2197 ETFDH KO cells was carried out as previously mentioned for three consecutive days. After 48h of recovery, cells were selected with 1 mg/mL hygromycin B for 3-4 days. ETFDH expression was confirmed by western blotting.

Generation of Cells Expressing EIF4EBP1

pBABE-puro-EV and pBABE-puro-4E-BP1 plasmids were used to overexpress 4E-BP1 in HCT-116 ETFDH KO cells. Retrovirus production and transduction was carried out similarly to lentivirus production, except 2.66 µg of pUMVC packaging plasmid (Addgene #8449) was utilized instead of psPAX2. Following three consecutive days of viral transduction, cells were allowed to recover for 48h prior to selection with 2 µg/mL puromycin (Sigma-Aldrich) for 3 days. To overexpress 4e-bp1 in NT2197 ETFDH KO cells, plasmids containing m4e-bp1 or ORF-Stuffer were generated by VectorBuilder (VB221206-1246sbx & VB900123-2599cba). Lentivirus production and transduction was carried out as previously mentioned for three consecutive days prior to a 48h recovery period. After this period, selection with hygromycin B (1 mg/mL) was conducted for 5-7 days. Validation was observed by western blotting.

Generation of Knockdown Models by shRNA/siRNA

Knockdown using shRNA was performed as previously described²⁰. For BCL6 knockdown experiments, the following shRNA vectors were used: pLKO.1 Non-Target shRNA Control (Sigma, SHC002), BCL6 shRNA #1 (Sigma, TRCN0000084654), or BCL6 shRNA #2 (TRCN0000084655). For

RAPTOR/RICTOR knockdown experiments, the following shRNA vectors were used: scramble shRNA (Addgene, Plasmid #1864), RAPTOR_1 shRNA (Addgene, Plasmid #1857), and RICTOR_1 shRNA (Addgene, Plasmid #1853). Briefly, HEK293T cells were transfected with psPAX2, pMD2.G or target pLKO.1 lentiviral shRNA vectors with JetPrime reagent (Polyplus) according to the manufacturer's instructions. Viral supernatants were filtered (0.45µM) and transferred to target cells with polybrene (6µg/mL). After 6 hours, cells were re-infected overnight. After 2 days, selection with puromycin (4µg/ml) was performed for 72 hours. Uninfected cells were selected with puromycin to serve as negative controls. Cells were seeded and processed for protein extraction or cell proliferation. Validation was confirmed by western blotting for *BCL6* knockdown and *RAPTOR* or *RICTOR* knockdown.

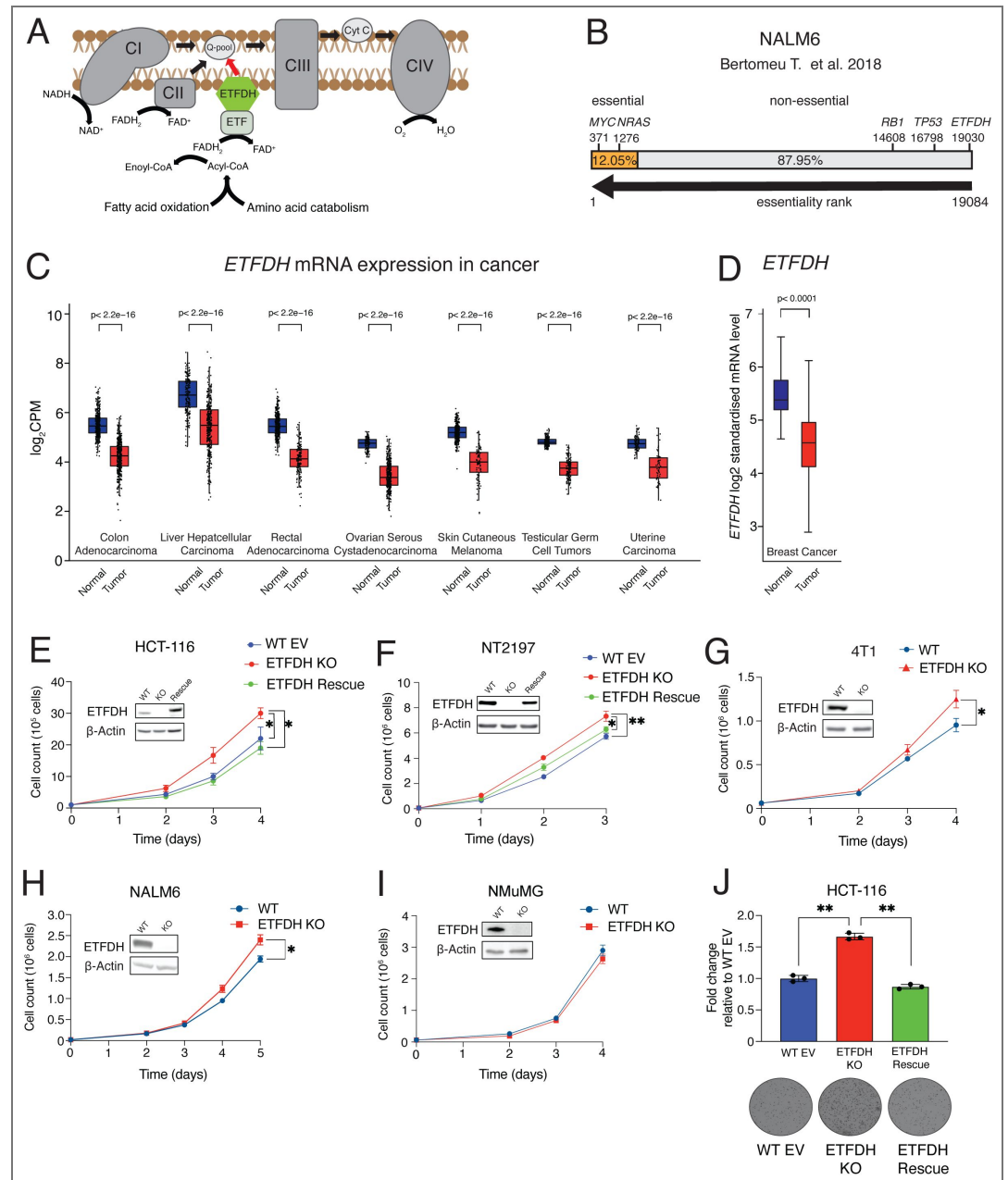
Knockdown using siRNA was performed as previously described⁶⁴. Briefly, target cells were transfected for 24 hours with 50nM Silencer Negative Control No.1 (Thermofisher, AM4611) or *BCL6* siRNA (Thermofisher, Cat#4390771) using JetPrime reagent (Polyplus) according to the manufacturer's instructions. Cells were re-transfected the next day for 24 hours, then processed for protein extraction. Validation was confirmed by western blotting.

Expression Profiles from TCGA/GTEx Databases

Expression of *ETFDH* and *EIF4EBP1* mRNA across both normal and tumor samples were obtained from The Cancer Genome Atlas (TCGA) and GTEx repositories. The expression data were normalized using the TMM (trimmed mean of M values) method and presented as log2 counts per million. Correlations between the genes were assessed using Pearson's correlation coefficient.

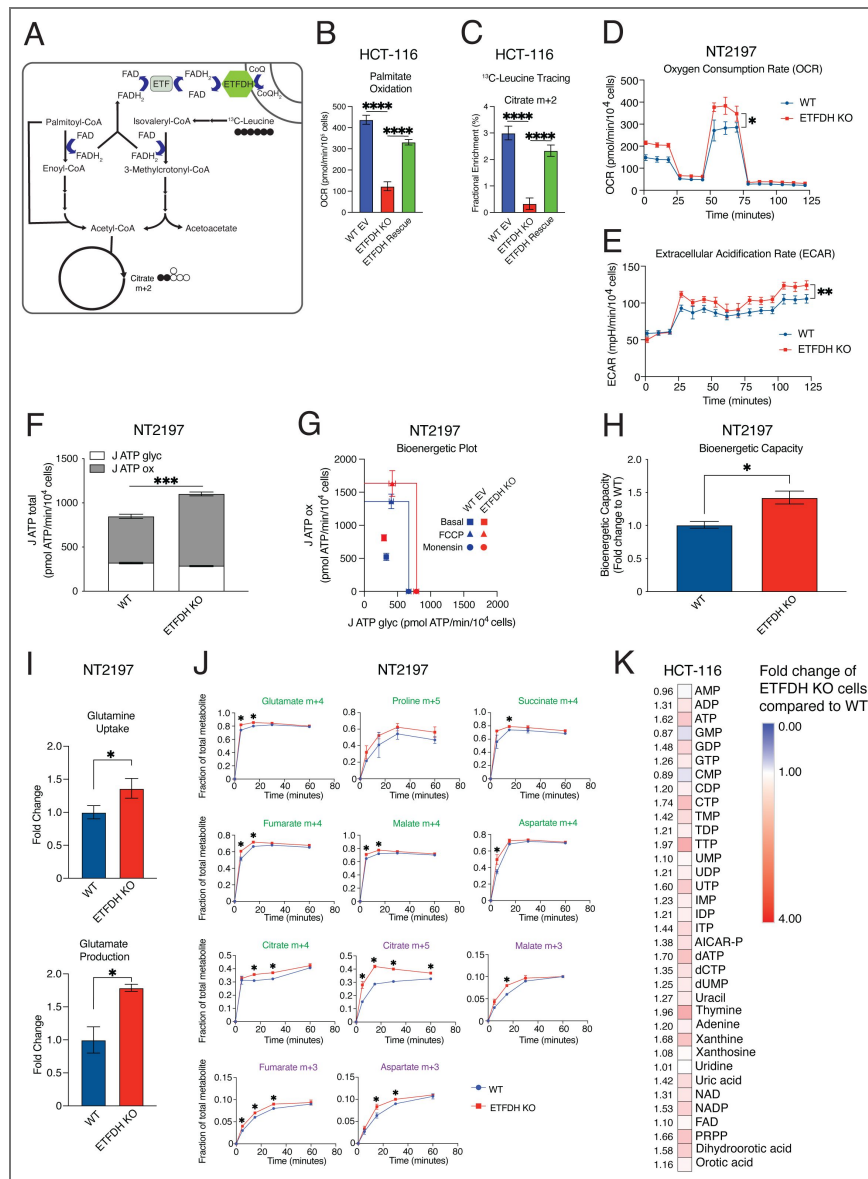
Quantification and Statistical Analysis

Statistical analysis was performed using GraphPad Prism 10. Data are presented as mean +/- SD of independent experiments, unless stated otherwise. Technical replicates were averaged from 2-3 independent experiments. Details on data quantification, presentation, and statistical analysis are included in figure legends.



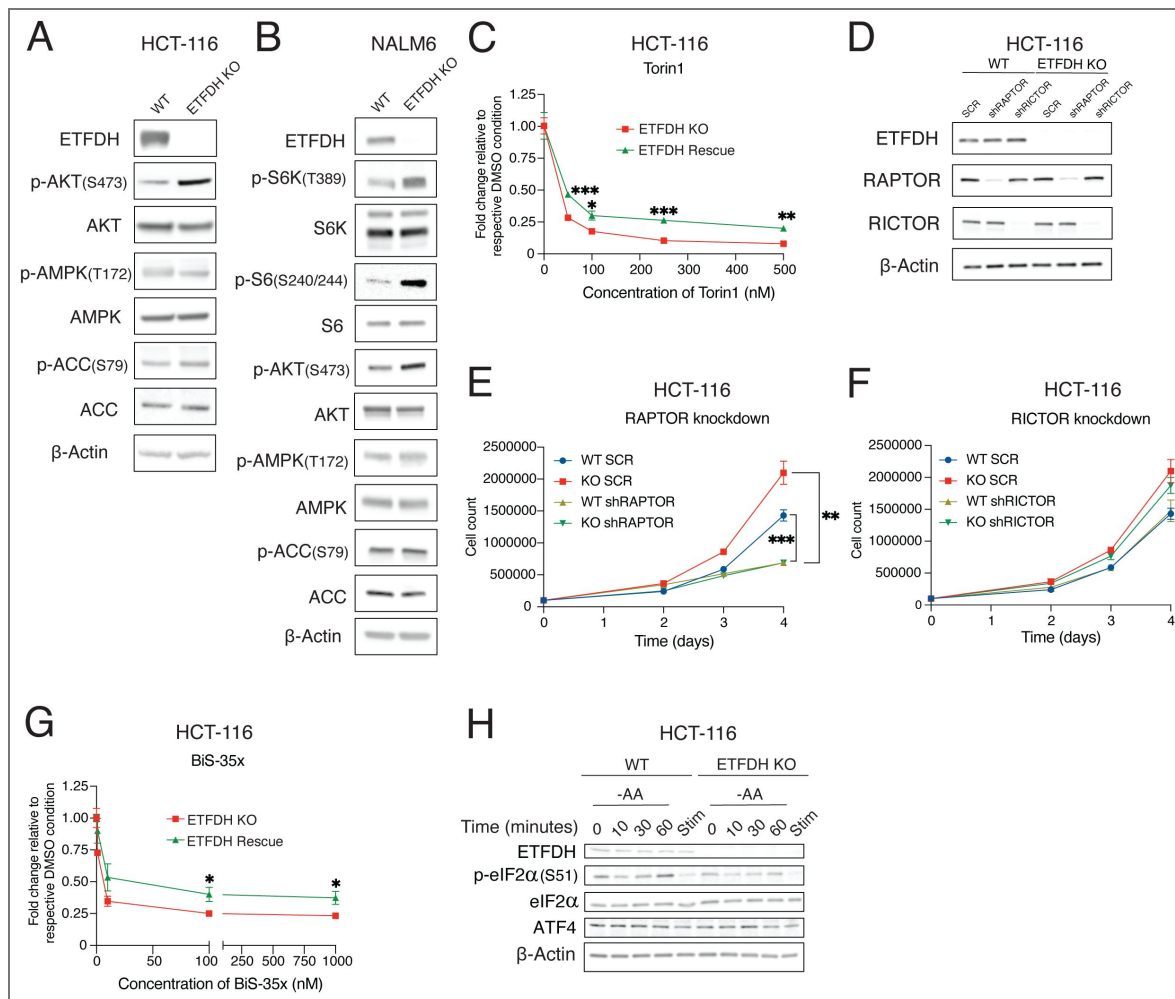
Supplementary Figure 1. *ETFDH* expression is reduced across a broad array of cancer subtypes, whereby *ETFDH* is a non-essential gene in cancer cells. A) Schematic of electron transfer from fatty and amino acids towards the electron transport chain (ETC) via electron transfer flavoprotein (ETF) and electron transport flavoprotein dehydrogenase (ETFDH). B) Gene essentiality in NALM6 cells. 19084 genes are presented and ranked based on essentiality (1 being most essential and 19084 being the least essential). Orange bar depicts essential genes and grey bar depicts non-essential genes. The rank of *ETFDH* is compared to those of *MYC*, *NRAS*, *RB1*, and *TP53*. C) *ETFDH* mRNA abundance profiles across tumor samples (red) compared to normal tissues (blue). Data are presented as log₂CPM (counts per million). Data were derived from The Cancer Genome Atlas (TCGA) repository and accessed on the GEPIA2 server (<http://gepia2.cancer-pku.cn>) (Colon Adenocarcinoma: Normal n=349, Tumor n=275; Liver Hepatocellular Carcinoma: Normal n=160, Tumor n=369; Rectal Adenocarcinoma: Normal n=318, Tumor n=92; Ovarian Serous Cystadenocarcinoma: Normal n=88, Tumor n=426; Skin Cutaneous Melanoma: Normal n=558, Tumor n=461; Testicular Germ Cell Tumors: Normal n=165, Tumor n=137; Uterine Carcinoma: Normal n=78, Tumor n=57), p<2.2x10⁻¹⁶. D) *ETFDH* mRNA levels across healthy (labelled as normal) and breast tumor (labelled as tumor) samples. RNAseq data from both TCGA and GTex databases were processed and normalized as described. TPM values were downloaded from GEO [GSE86354 (healthy) and [GSM1536837 (tumor)], p<0.0001. E) HCT-116. Cell growth curves for WT EV (blue), ETFDH KO (red), and ETFDH Rescue (green) cells. Data are presented as cell count (10³ cells) over time (days). WT EV and ETFDH Rescue cells show similar growth, while ETFDH KO cells show significantly reduced growth (*). F) NT2197. Cell growth curves for WT EV (blue), ETFDH KO (red), and ETFDH Rescue (green) cells. Data are presented as cell count (10³ cells) over time (days). WT EV and ETFDH Rescue cells show similar growth, while ETFDH KO cells show significantly reduced growth (**). G) 4T1. Cell growth curves for WT (blue) and ETFDH KO (red) cells. Data are presented as cell count (10³ cells) over time (days). WT and ETFDH KO cells show similar growth. H) NALM6. Cell growth curves for WT (blue) and ETFDH KO (red) cells. Data are presented as cell count (10³ cells) over time (days). WT and ETFDH KO cells show similar growth. I) NMuMG. Cell growth curves for WT (blue) and ETFDH KO (red) cells. Data are presented as cell count (10³ cells) over time (days). WT and ETFDH KO cells show similar growth. J) HCT-116. Fold change relative to WT EV for WT EV, ETFDH KO, and ETFDH Rescue cells. Data are presented as fold change relative to WT EV. WT EV and ETFDH Rescue cells show similar growth, while ETFDH KO cells show significantly reduced growth (**).

(tumour)]. Expression is presented as $\log_2$ standardised mRNA levels. Data were acquired from the 'Breast cancer gene-expression miner' (bc-GenExMiner) server (<http://bcgenex.ico.unicancer.fr>) using *ETFDH* expression according to nature of the tissue. (Normal n=92, Tumor n=743). $p < 0.0001$, Dunnett-Tukey-Kramer's test. E-F) Proliferation of WT EV, ETFDH KO, and ETFDH Rescue HCT-116 (E) and NT2197 (F) cells. ETFDH rescue measurements are the same as in Fig. 6I. Data are presented as cell count means $\pm$ SD over multiple time-points. Representative western blot of indicated proteins is provided in the insets (n=3), * $p < 0.05$, ** $p < 0.01$, one-way ANOVA, Tukey's post hoc test. G-H) Proliferation of WT and ETFDH KO 4T1 (G) and NALM6 (H) cells. Data are presented as cell count means $\pm$ SD over indicated time-points. Representative western blot of indicated proteins is provided in the insets (n=3), * $p < 0.05$, paired Student's *t* test. I) Proliferation of WT and ETFDH KO NMuMG cells. Representative western blot of indicated proteins is provided in the insets (n=3), one-way ANOVA, Dunnett's post hoc test. J) Soft agar/colony formation assays in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells. Data are presented as fold change relative to WT EV cells (n=3), * $p < 0.05$, ** $p < 0.01$, one-way ANOVA, Tukey's post-hoc test.



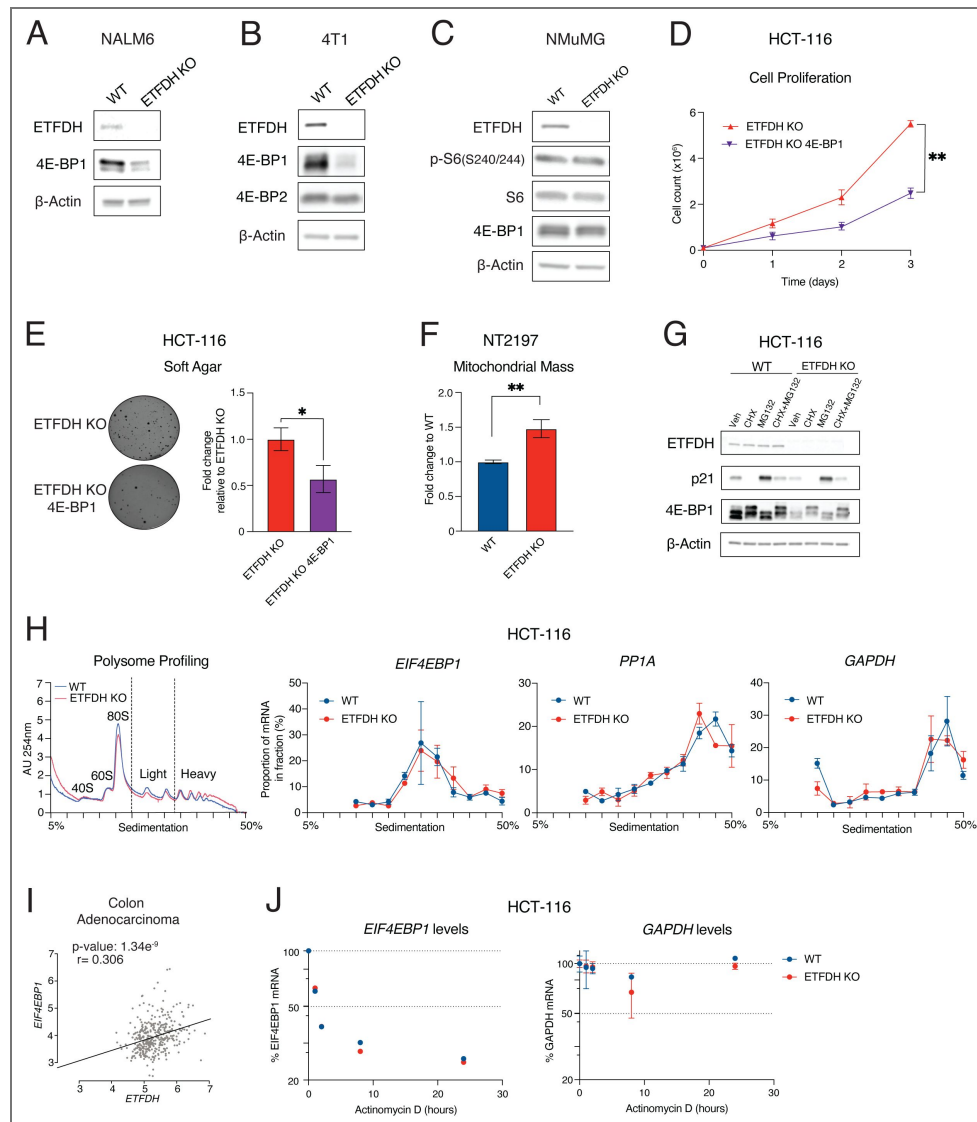
Supplementary Figure 2. ETFDH loss reprograms mitochondrial bioenergetics

A) Schematic depicting fatty acid and leucine catabolism linked to ETFDH electron transfer. B) Palmitate oxidation at basal level in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells. Data are presented as means $\pm$ SD ($n=6$), **** $p<0.0001$, one-way ANOVA, Tukey's post-hoc test. C) Fractional enrichment (%) of citrate m+2 in WT EV, ETFDH KO, and ETFDH Rescue HCT-116 cells. Data are presented as means $\pm$ SD ($n=6$), **** $p<0.0001$, one-way ANOVA, Tukey's post-hoc test. D-E) Oxygen consumption (D) and extracellular acidification (E) for WT and ETFDH KO NT2197 cells was determined using a Seahorse bioanalyzer. Data are normalized to cell count and presented as means $\pm$ SD ($n=5$), * $p<0.05$, ** $p<0.01$, paired Student's t test. F) Basal J ATP calculations from WT and ETFDH KO NT2197 cells. J ATP ox represents ATP production from oxidative phosphorylation, while J ATP glycol represents ATP production from glycolysis. Comparison between J ATP ox (grey bars; top) and J ATP glycol (white bars; bottom) is shown. Data are presented as means $\pm$ SD ($n=5$), *** $p<0.001$, paired Student's t test. G-H) Bioenergetic Plot for Basal, FCCP, and Monensin J ATP fluxes (G) and Bioenergetic Capacity (H) for WT and ETFDH KO NT2197 cells. Data are presented as means $\pm$ SD ($n=5$), * $p<0.05$, paired Student's t test. I) Glutamine uptake and glutamate production in WT and ETFDH KO NT2197 cells. Data are presented as mean fold changes of ETFDH KO cells relative to WT cells $\pm$ SD ($n=3$), * $p<0.05$, paired Student's t test. J) Fractional abundance of ¹³C-labelled metabolites over indicated timepoints from WT (blue) and ETFDH KO (red) NT2197 cells. Isotopomers labelled in green depict ¹³C-glutamine tracing in the forward direction of the CAC, while those in purple represent reverse tracing. Data are presented as means $\pm$ SD ($n=3$), * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$, two-way ANOVA, Tukey's post hoc test. K) Nucleotide metabolite profile in WT and ETFDH KO HCT-116 cells monitored by LC-MS/MS. Data are presented as fold change in ETFDH KO compared to WT cells ($n=3$).



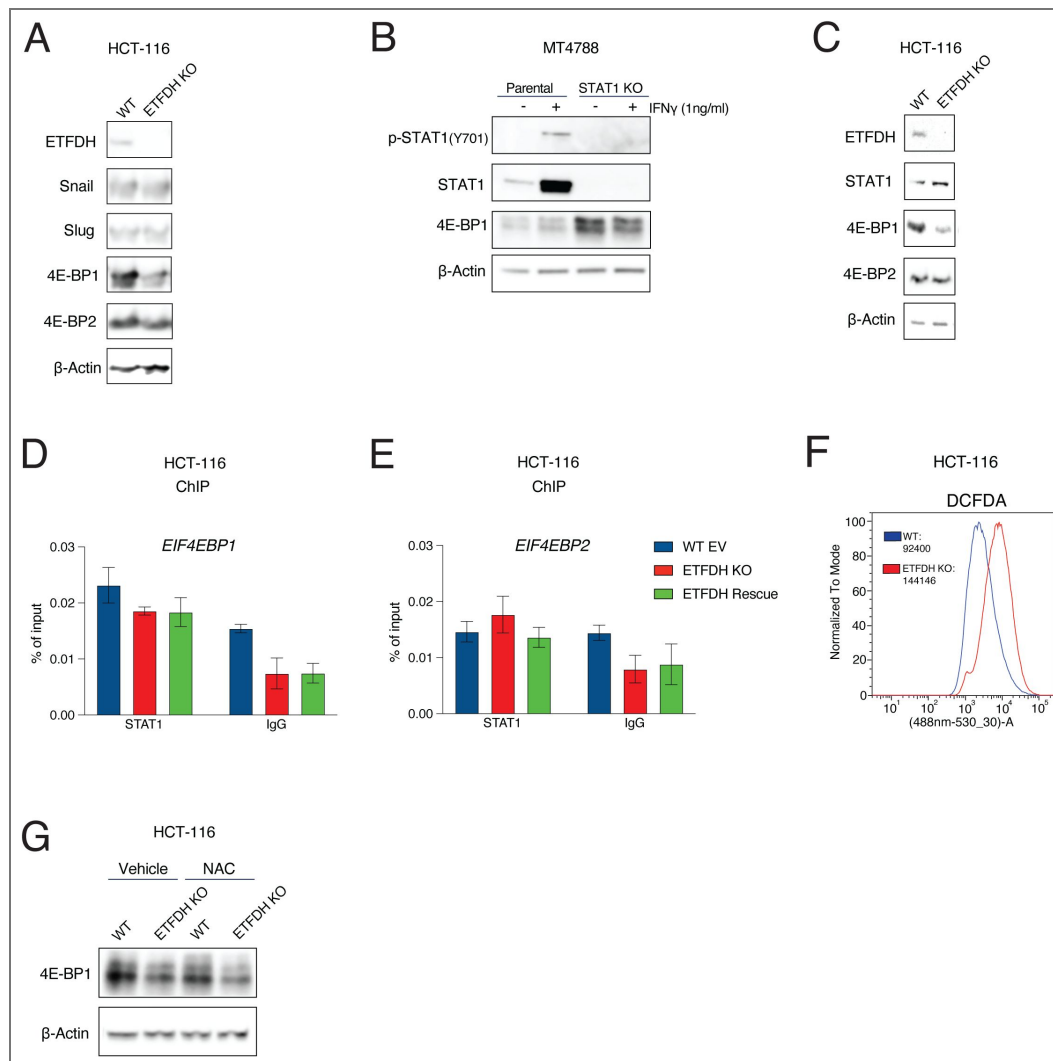
Supplementary Figure 3. ETFDH loss increases mTORC1 signaling and dependency

A-B) Levels and phosphorylation status of indicated proteins in WT and ETFDH KO HCT-116 (A) and NALM6 (B) cells were determined by western blotting. β-Actin was used as a loading control (representative blots of n=3). C) Proliferation of HCT116 ETFDH KO and ETFDH Rescue cells treated with indicated concentrations of Torin1 for 72h. Data are presented as fold changes relative to vehicle (DMSO) treatment (n=3), *p<0.05, **p<0.01, ***p<0.001, two-way ANOVA, Tukey's post hoc test. D) Indicated protein levels in WT and ETFDH KO HCT-116 cells transfected with shRNA targeting *RAPTOR* (shRAPTOR), *RICTOR* (shRICTOR), or scrambled control (SCR) were determined by western blotting. β-Actin was used as a loading control (representative blots of n=3). E-F) Proliferation of HCT-116 WT and ETFDH KO cells expressing SCR and shRAPTOR (E) or SCR and shRICTOR (F). Data are presented as cell count means ± SD (n=4), **p<0.01, ***p<0.001 two-way ANOVA, Tukey's post hoc test. G) Proliferation of HCT116 ETFDH KO and ETFDH Rescue cells treated with indicated concentrations of Bis-35x for 72 hours. Data are presented as fold changes relative to vehicle (DMSO) treatment (n=3), *p<0.05, **p<0.01, ***p<0.001, two-way ANOVA, Tukey's post hoc test. H) Total and phosphoprotein levels in WT or ETFDH KO HCT-116 cells were determined by western blotting using indicated proteins. Cells were serum starved overnight, then depleted of amino acids for indicated timepoints (0, 10, 30, 60 minutes), or stimulated with FBS for 4 hours (Stim). β-Actin was used as loading control (representative blots of n=3).



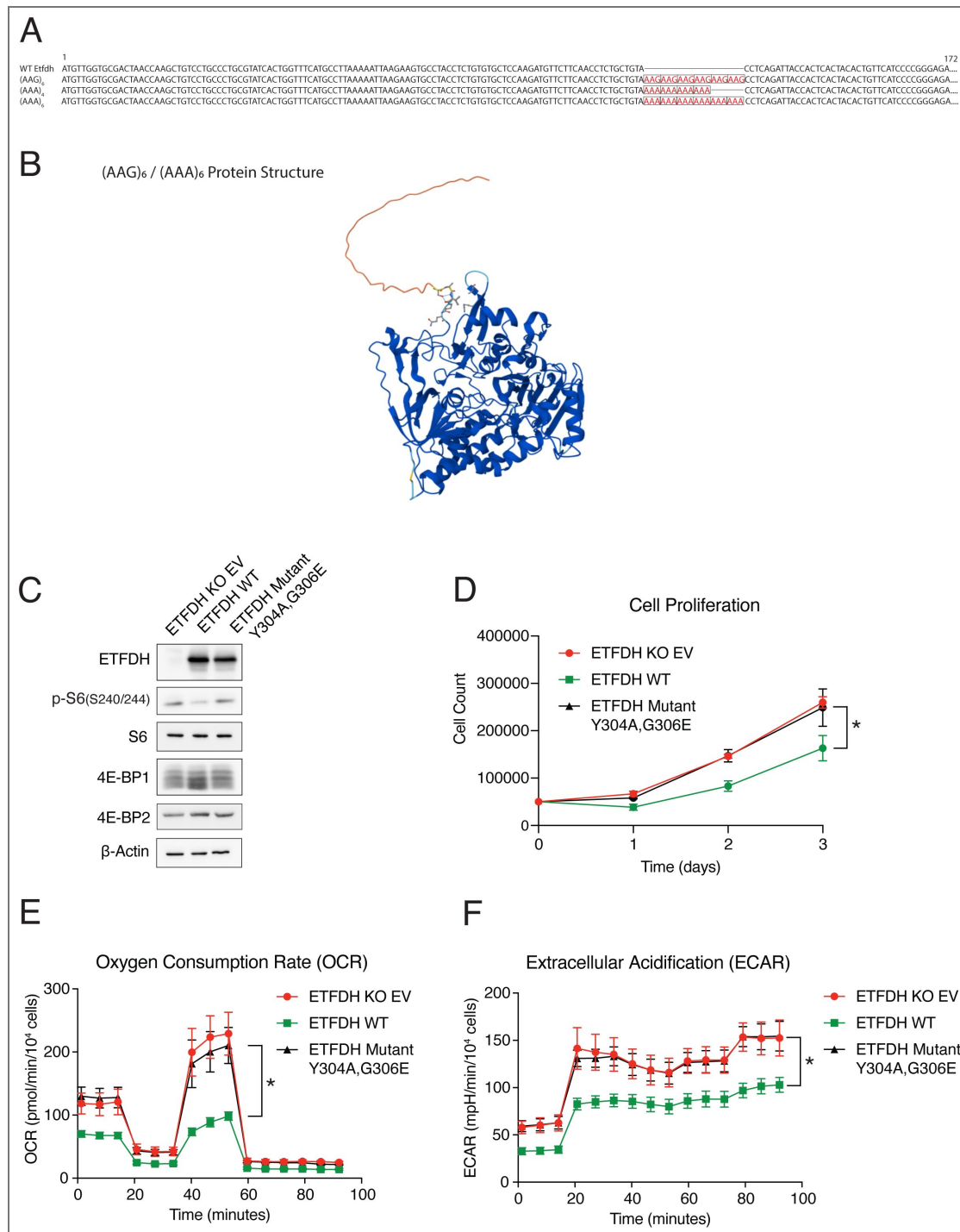
Supplementary Figure 4. *ETFDH* loss induces *EIF4EBP1* transcription

A) Protein levels in WT and ETFDH KO NALM6 cells were monitored by western blotting using indicated antibodies. β -Actin was used as a loading control (representative blots of $n=3$). B) Indicated protein levels in WT and ETFDH KO 4T1 cells were determined by western blotting using indicated antibodies. β -Actin was used as a loading control (representative blots of $n=3$). C) Protein levels in WT and ETFDH KO NMuMG cells were monitored by western blotting using indicated antibodies. β -Actin was used as a loading control (representative blots of $n=3$). D) Proliferation of ETFDH KO and ETFDH KO overexpressing 4E-BP1 (ETFDH KO 4E-BP1) HCT-116 cells. Data are presented as cell count means $\pm$ SD over indicated time-points ($n=3$), ** $p < 0.01$, paired Student's t test. E) Soft agar/colony formation assay in ETFDH KO and ETFDH KO 4E-BP1 HCT-116 cells. Data are presented as fold change relative to ETFDH KO cells ($n=3$), * $p < 0.05$, paired Student's t test. F) Mitochondrial mass in WT and ETFDH KO NT2197 cells was monitored by flow cytometry. Data are presented as fold change relative to WT cells ($n=3$), ** $p < 0.01$, paired Student's t test. G) The levels of indicated proteins in WT and ETFDH KO HCT-116 cells treated with cycloheximide (CHX) (50 μ g/ml), MG-132 (10 μ M), or combination thereof for 4 hours total were monitored by western blotting. β -Actin was used as loading control, whereas p21 was used as a short-lived protein to control for the effects of the CHX and MG-132 treatments (representative blots of $n=2$). H) Polysome profiling of WT and ETFDH HCT-116 KO cells. Absorbance at 254nm (AU 254nm) was used to monitor the distribution of 40S-, 60S-ribosomal subunits, monosomes (80S) and polysomes across the gradient. Indicated mRNA abundances from polysome gradient fractions was obtained using RT-qPCR. Data are shown as a mean percentage of mRNA in each fraction relative to cumulative corresponding mRNA amount across the whole gradient $\pm$ SD ($n=3$). I) Correlation between *ETFDH* and *EIF4EBP1* mRNA expression in colon adenocarcinoma samples. Data were derived from TCGA repository. J) WT and ETFDH KO HCT-116 cells were treated with Actinomycin D (10 μ g/ml) for indicated timepoints upon which *EIF4EBP1* and *GAPDH* mRNA levels were determined by RT-qPCR. Data are presented as a mean percentage of mRNA relative to DMSO-treated cells $\pm$ SD ($n=3$).



Supplementary Figure 5. The effects of ETFDH loss on *EIF4EBP1* transcription are mediated by BCL-6 but not STAT1, Snail or Slug

A) Protein levels in WT and ETFDH KO HCT-116 cells were determined by western blotting using indicated antibodies. β -Actin was used as a loading control (representative blots of n=2). B) MT4788 WT and STAT1 KO cells were treated with IFN γ (1ng/ml) to induce STAT1 or vehicle (water) for 24 hours. Levels and phosphorylation status of indicated proteins were determined by western blotting. β -Actin was used as a loading control (representative blots of n=2). C) Protein levels in WT and ETFDH KO HCT-116 cells were assessed by western blotting using annotated antibodies. β -Actin served as a loading control (representative blots of n=3). D-E) Binding of STAT1 to the promoters of *EIF4EBP1* (D) and *EIF4EBP2* (E) was determined by ChIP-qPCR. IgG was used a negative control. Data are presented as a percentage of the input $\pm$ SD (n=3), one-way ANOVA, Tukey's post-hoc test. F) Fluorescence intensity of WT and ETFDH KO HCT-116 cells stained with DCFDA (2 μ M) and measured by flow cytometry. Median values are depicted in the inlets. G) Levels of 4E-BP1 in WT and ETFDH KO HCT-116 cells were determined by western blotting using the indicated antibody. β -Actin was used as a loading control (representative blots of n=3).



Supplementary Figure 6. Poly(A) Track design (A-B); ETFDH catalytic activity is required for its tumor suppressive function (C-F)

A) Insertion of poly(A) tracks within the coding sequence of murine ETFDH. DNA sequences for constructs (AAG)₆, (AAA)₄, (AAA)₆, and WT Etfhd are shown. B) Protein structure of (AAG)₆ / (AAA)₆ constructs generated using AlphaFold. C) Levels of indicated proteins in ETFDH KO EV, WT ETFDH rescue (ETFDH WT), and ETFDH mutant Y304A, G306E HCT-116 cells were determined by western blotting. β-Actin was used as a loading control (representative blots of n=2). D) Proliferation of ETFDH KO EV, ETFDH WT, and ETFDH mutant Y304A, G306E HCT-116 cells. Data are presented as cell count means ± SD over indicated time-points (n=3), *p<0.05, one-way ANOVA, Tukey's post-hoc test. E-F) Oxygen consumption (E) and extracellular acidification (F) for ETFDH KO EV, ETFDH WT, and ETFDH mutant Y304A, G306E HCT-116 cells was determined using a Seahorse bioanalyzer. Data are normalized to cell count and presented as means ± SD (n=4), *p<0.05, one-way ANOVA, Tukey's post-hoc test.

Supplementary Table 1. List of Antibodies

Antibody	Source	Identifier
Mouse monoclonal β -actin antibody, 1:5000 dilution	MilliporeSigma	Cat#A5441; RRID: AB_476744
Normal Rabbit IgG	Cell Signaling Technologies	Cat#2729; RRID: AB_1031062
Rabbit monoclonal 4E-BP1 (53H11) antibody, 1:1000 dilution	Cell Signaling Technologies	Cat#9644; RRID: AB_2097841
Rabbit monoclonal Akt (pan) (C67E7) antibody, 1:1000 dilution	Cell Signaling Technologies	Cat#4691; RRID: AB_915783
Rabbit monoclonal phospho-4E-BP1 (S65) (174A9) antibody, 1:1000 dilution	Cell Signaling Technologies	Cat#9456; RRID: AB_823413
Rabbit monoclonal Phospho-Akt (Ser473) (D9E) XP® antibody, 1:1000 dilution	Cell Signaling Technologies	Cat#4060; RRID: AB_2315049
Rabbit monoclonal Rictor (53A2) antibody, 1:1000 dilution	Cell Signaling Technologies	Cat#2114; RRID: AB_2179963
Rabbit polyclonal phospho-S6 Ribosomal Protein (Ser240/244) antibody, 1:2000 dilution	Cell Signaling Technologies	Cat#2215; RRID: AB_2630325
Rabbit polyclonal Raptor antibody, 1:1000 dilution	MilliporeSigma	Cat#09-217; RRID: AB_612103
Rabbit polyclonal 4E-BP2 antibody, 1:1000 dilution	Cell Signaling Technology	2845; RRID: AB_10699019
Rabbit monoclonal phospho-S6K (T389) antibody, 1:500 dilution	Cell Signaling Technology	9234; RRID: AB_2269803
Rabbit monoclonal S6K antibody, 1:1000 dilution	Cell Signaling Technology	2708; RRID: AB_390722
Mouse monoclonal S6 ribosomal protein antibody, 1:5000 dilution	Santa Cruz Biotechnology	sc-74459; RRID: AB_1129205
Rabbit polyclonal ACC antibody, 1:1000 dilution	Cell Signaling Technology	3662; RRID: AB_2219400
Rabbit polyclonal phospho-ACC (Ser79) antibody, 1:1000 dilution	Cell Signaling Technology	3661; RRID: AB_330337
Mouse monoclonal eIF2 α (L57A5) antibody, 1:1000 dilution	Cell Signaling Technology	2103; RRID: AB_836874
Rabbit polyclonal eIF2 α (Ser51) antibody, 1:1000 dilution	Cell Signaling Technology	9721; RRID: AB_330951
Rabbit monoclonal eIF4G1 antibody, 1:500 dilution	Cell Signaling Technology	2858; RRID: AB_2095745
Rabbit polyclonal TFAM antibody, 1:1000 dilution	Cell Signaling Technology	7495; RRID: AB_10841294
Rabbit monoclonal GAPDH (14C10) antibody, 1:1000 dilution	Cell Signaling Technology	2118; RRID: AB_561053
Mouse monoclonal eIF4E antibody, 1:1000 dilution	BD Biosciences	610269; RRID: AB_397664
Mouse monoclonal puromycin (12D10) antibody, 1:1000 dilution	Millipore	MABE343; RRID: AB_2566826
Rabbit polyclonal AMPK α antibody, 1:500 dilution	Cell Signaling Technology	2532; RRID: AB_330331
Rabbit monoclonal Anti-AMPK- α , phospho (Thr172) antibody, 1:500 dilution	Cell Signaling Technology	2535; RRID: AB_331250
Rabbit polyclonal ETFDH antibody, 1:2000 dilution	ProteinTech	11109-1-AP; RRID: AB_2231382
Rabbit monoclonal BCL-6 antibody, 1:1000 dilution	Cell Signaling Technology	14895; RRID: AB_2798638
Rabbit monoclonal BCL-6 antibody, 1:1000 dilution	Cell Signaling Technology	5650; RRID: AB_10949970
Rabbit monoclonal Snail antibody, 1:1000 dilution	Cell Signaling Technology	3879; RRID: AB_2255011
Rabbit monoclonal Slug antibody, 1:1000 dilution	Cell Signaling Technology	9585; RRID: AB_2239535
Rabbit polyclonal CREB-2 (ATF4) antibody, 1:1000 dilution	Santa Cruz	sc-200; RRID: AB_2058752
Rabbit monoclonal STAT1 antibody, 1:1000 dilution	Cell Signaling Technology	14994; RRID: AB_2737027
Rabbit monoclonal p-STAT1 (Tyr701) antibody, 1:1000 dilution	Cell Signaling Technology	9167; RRID: AB_561284

Supplementary Table 2. Primer sequences used for RT-Qpcr

Primer Name	Primer Sequence
Human	
EIF4EBP1 forward	GGAAGTCACCTGTGACCAAAA
EIF4EBP1 reverse	CCGCTTATCTTCTGGGCTATT
EIF4EBP2 forward	AGTCACTAGCCCTGGCACCTTA
EIF4EBP2 reverse	CTTGACGAGAGTGCAGATGTCC
BCL6 forward	TGAGAAAGCCCTATCCCTGTG
BCL6 reverse	GTGACGCGAAATGCAGGTTA
PP1A forward	GTCAACCCACCGTGTCTT
PP1A reverse	CTGCTGTCTTTGGGACCTTGT
GAPDH forward	TGTTGCCATCAATGACCCCTT
GAPDH reverse	CTCCACGACGTACTACGCG
Mouse	
Eif4ebp1 forward	GGTCACTAGCCCTACCAGCG
Eif4ebp1 reverse	TTGTGACTCTTACCGCCTG
Eif4ebp2 forward	TACCTCAGGACTACTGCACCAC
Eif4ebp2 reverse	GGAGAATTGCGACGGTCCAACA
Bcl6 forward	GCGAACCTTGATCTCCAGTC
Bcl6 reverse	TGACTCTCACTGCTGCTTCG
Actb forward	TTCTTCTTGGGTATGGAA
Actb reverse	CCACGATCCACACAGATA

Supplementary Table 3. Primer sequences used for ChIP-qPCR

Primer Name	Primer Sequence
EIF4EBP1 ChIP forward	TGACTGAGTTTCGGATCTGG
EIF4EBP1 ChIP reverse	CACTCGCGTAAATGGGGAAAC
EIF4EBP2 ChIP forward	TTCAACAACCTTACGCGACGC
EIF4EBP2 ChIP reverse	GGAGTTGGGGAAGCTGGTC

Supplementary Table 4. gRNA sequences used for generation of ETFDH KO cell lines

Primer Name	Primer Sequence
Human	
ETFDH gRNA Target 1 Forward	CACCGAGTTGGCCGAATGCTAGGA
ETFDH gRNA Target 1 Reverse	AAACTCTAGCATTCCGCCAACCTC
ETFDH gRNA Target 2 Forward	CACCGATGAGGGATACAAAAGGA
ETFDH gRNA Target 2 Reverse	AAACTCCTTTGTATCCCTACATC
Mouse	
ETFDH gRNA Target 1 Forward	CACCGGTGATACGCTGCATAAAAT
ETFDH gRNA Target 1 Reverse	AAACATTTTATGCAGCGTATCACC
ETFDH gRNA Target 2 Forward	CACCGGAACATCTTGGAGCACACAG
ETFDH gRNA Target 2 Reverse	AAACCTGTGTCTCCAAGATGTTCC

Primer Name	Primer Sequence
Human	
ETFDH validation Set 1 Forward	AGCGAGCATTTTCCTCCACA
ETFDH validation Set 1 Reverse	ACCATTGTTAGCTCAAAGTTGCT
ETFDH validation Set 2 Forward	TCATTCCCATTGTGCGAGC
ETFDH validation Set 2 Reverse	ATGTACAGTTTGTAGCAAGACCT
Mouse	
ETFDH validation Set 1 Forward	GTGTCTGAAGGCAGCAACAG
ETFDH validation Set 1 Reverse	GGATCAAGACAAGCCCTGA
ETFDH validation Set 2 Forward	CCAGGTTCAATCCCAGCAC
ETFDH validation Set 2 Reverse	CCGGGGATGAACAGTGTAGT

Supplementary Table 5. Primer sequences used for sequencing

Data availability

Plasmids generated in this study is deposited on Addgene. The original data is available at Mendeley Data (doi: 10.17632/p4wxkr3vdm.1). Metabolomics data is available at MetaboLights (MTBLS11048).

Acknowledgements

We are thankful to the members of all involved laboratories and Stephane Richard for helpful discussions. We thank Luc Choinière and Mariana Russo for assistance with metabolomic experiments, Christian Young for assistance with flow cytometry, and Christopher Rudd for assistance with respirometry. This research was funded by the Terry Fox Foundation (TFF) Oncometabolism Team Grant (TFF-242122) to I.T., P.S, and M.P, and Canadian Institutes of Health Research (CIHR) (PJT-183843, PJT-479494) to I.T. Research in OL's lab is supported by grants from the Swedish Research Council (2020-01665), Swedish Cancer Society (22 2186), the Cancer Research Funds of Radiumhemmet (231263) and the Wallenberg Academy Fellow Program. L-M.P. acknowledges the funds from Queen's University. D.P. is supported by CIHR Postdoctoral Fellowship (MFE-171312), Cancer Research Society (CRS) The Next Generation of Scientists Award (NGS), and LDI Desjardins Fellowship Program. P.J. and H.K. are supported by Fonds de Recherche du Québec – Santé (FRQS) Fellowships. I.T. is supported by Canada Research Chair in Regulation of mRNA Translation and Metabolism. Metabolic analysis was performed at the Rosalind and Morris Goodman Cancer Research Centre's Metabolomics Core Facility, which is supported by the Canada Foundation for Innovation, the Dr. John R. and Clara M. Fraser Memorial Trust, the Terry Fox Foundation (TFF Oncometabolism Team Grant; TFF-242122), and McGill University.

Additional information

Author Contributions

Conceptualization, D.P., M.P. and I.T.; Methodology, D.P., R.P., P.J., S.I., H.K., T.B., and S.D.; Formal Analysis, D.P., P.J., S.T., E.C., S.I., K.J.S. and T.B.; Investigation, D.P., R.P., P.J., S.T., C., V.S., S.I., S.M., L.Z., H.K., N.C., K.J.S., J.Z., J.V., F.A., and S.M.; Resources, E.G., M.T, D.A., O.L., L.-M.P., S.D., J.U.-S., P.M.S., M.P., and I.T.; Writing – Original Draft, D.P., M.P., and I.T.; Writing – Review & Editing, all authors; Supervision, M.P., and I.T.; Project Administration, M.P., and I.T.; Funding Acquisition, D.P., M.P., and I.T.

Material & Correspondence

Further information and requests for reagents may be directed upon reasonable request by the lead contact, Ivan Topisirovic (ivan.topisirovic@mcgill.ca [↗](#)).

Funding

Funder	Grant reference number
Terry Fox Foundation	TFF-242122
Canadian Institutes of Health Research	PJT-183843
Canadian Institutes of Health Research	PJT-479494
Canadian Institutes of Health Research	MFE-171312
Swedish Research Council	2020-01665
Swedish Cancer Society	22 2186

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Peer reviews

Reviewer #1 (Public review):

In their manuscript, Papadopoli et al explore the role of ETFDH in transformation. They note that ETFDH protein levels are decreased in cancer, and that deletion of ETFDH in cancer cell lines results in increased tumorigenesis, elevated OXPHOS and glycolysis, and a reduction in lipid and amino acid oxidation. The authors attribute these effects to increased amino acid levels stimulating mTORC1 signaling and driving alterations in BCL6 and EIF4EBP1. They conclude that ETFDH1 is epigenetically silenced in a proportion of neoplasms, suggesting a tumor-suppressive function. Overall, the authors logically present clear data and perform appropriate experiments to support their hypotheses.

<https://doi.org/10.7554/eLife.106587.2.sa2>

Reviewer #2 (Public review):

Summary:

The altered metabolism of tumors enables their growth and survival. Classically, tumor metabolism often involves increased activity of a given pathway in intermediary metabolism to provide energy or substrates needed for growth. Papadopoli et al. investigate the converse - the role of mitochondrial electron transfer flavoprotein dehydrogenase (ETFDH) in cancer metabolism and growth. The authors present compelling evidence that ETFDH insufficiency, which is detrimental in non-malignant tissues, paradoxically enhances bioenergetic capacity and accelerates neoplastic growth in cancer cells in spite of the decreased metabolic fuel flexibility that this affords tumor cells. This is achieved through the retrograde activation of the mTORC1/BCL-6/4E-BP1 axis, leading to metabolic and signaling reprogramming that favors tumor progression.

Strengths:

This review focuses primarily on the cancer metabolism aspects of the manuscript.

The study provides robust evidence linking ETFDH insufficiency to enhanced cancer cell bioenergetics and tumor growth.

The use of multiple cancer cell lines and in vivo models strengthens the generalizability of the findings.

The mechanistic insights into the mTORC1/BCL-6/4E-BP1 axis and its role in metabolic reprogramming are of general interest within and outside the immediate field of tumor metabolism.

Conclusion:

This manuscript provides significant insights into the role of ETFDH insufficiency in cancer metabolism and growth. The findings highlight the potential of targeting the mTORC1/BCL-6/4E-BP1 axis in ETFDH-deficient cancers. The compelling data support the conclusions presented in the manuscript, which will be valuable to the cancer metabolism community.

[Editors' note: The authors have addressed each of the two weaknesses previously listed in the public review, providing new experimental data on nucleotides and showing that the catalytic activity is required via the suggested addback experiment.]

<https://doi.org/10.7554/eLife.106587.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Authors state, "we identified ETF dehydrogenase (ETFDH) as one of the most dispensable metabolic genes in neoplasia." Surely there are thousands of genes that are dispensable for neoplasia. Perhaps the authors can revise this sentence and similar sentiments in the text.

We agree with the reviewer and have corrected the text accordingly. Specifically, we rephrased the sentence: "Surprisingly, we observed that in contrast to muscle, ETFDH is one of the most non-essential metabolic genes in cancer cells." to "Surprisingly, we observed that

in contrast to muscle, ETFDH is a non-essential gene in acute lymphoblastic leukemia NALM-6 cells”

Authors state, "These findings show that ETFDH loss elevates glutamine utilization in the CAC to support mitochondrial metabolism." While elevated glutamine to CAC flux is consistent with the statement that increased glutamine, the authors have not measured the effect of restoring glutamine utilization to baseline on mitochondrial metabolism. Thus, the causality implied by the authors can only be inferred based on the data presented. Indeed, the increased glutamine consumption may be linked to the increase in ROS, as glutamate efflux via system xCT is a major determinant of glutamine catabolism in vitro.

Indeed. We changed the statement "These findings show that ETFDH loss elevates glutamine utilization in the CAC to support mitochondrial metabolism." to "Collectively, these data demonstrate that ETF insufficiency in cancer cells remodels mitochondrial metabolism and increases the glutamine consumption and anaplerosis."

Authors state that the mechanism described is an example of "retrograde signaling". However, the mechanism seems to be related to a reduction in BCAA catabolism, suggesting that the observed effects may be a consequence of altered metabolic flux rather than a direct signaling pathway. The data presented do not delineate whether the observed effects stem from disrupted mitochondrial communication or from shifts in nutrient availability and metabolic regulation.

Notwithstanding that the term “retrograde” was used to refer to signaling from mitochondria to mTORC1, rather than from mTORC1 to mitochondria [1], we have removed the term “retrograde signaling” throughout the manuscript.

The authors should discuss which amino acids that are ETFDH substrates might affect mTORC1 activity or consider whether other ETFDH substrates might also affect mTORC1 in their discussion. Along these lines, the authors might consider discussing why amino acids that are not ETFDH substrates are increased upon ETFDH loss.

Based on the literature, we expect that branched chain amino acids that are ETFDH substrates (e.g., leucine) are likely to play a major role in activating mTORC1 upon ETFDH abrogation. As expected, the aforementioned amino acids are among those that are the most highly upregulated in ETFDH deficient cells (Fig 3A). We have, however, never formally tested the role of branched chain amino acid in activating mTORC1 in the context of ETFDH disruption. The increase in amino acids that are not metabolized via ETFDH, is likely to stem from global metabolic rewiring of ETFDH-deficient cells and observed alterations in amino acid uptake (e.g., glutamine; Fig 2F). We discuss this in the revised version of the paper as follows:

“Several metabolites can be sensed via signaling partners upstream of mTORC1, including leucine, arginine, methionine/SAM, and threonine [2]. Branched-chain amino acids (leucine, isoleucine, and valine), which are among the highest upregulated metabolites in ETFDH deficient cells (Fig 3A) serve as ETFDH substrates, and have been described to display strong activation capabilities towards mTORC1 in the literature [3,4]. Glutamine can also activate mTORC1 through Arf family of GTPases [5]. Indeed, glutamine can supplement the non-essential amino acid (NEAA) pool through transamination [6] and amino acid uptake [7]. Accordingly, the maintenance of NEAA that are non-ETFDH substrates may be supported by the global metabolic rewiring fueled by enhanced glutamine metabolism in ETFDH-deficient cells. Deciphering the mechanisms leading to accumulation of specific amino acids and their role in ETFDH-dependent mTORC1 modulation is warranted.”

Reviewer #2 (Public review):

The authors would strengthen the paper considerably by adding back catalytically inactive ETFDH to show that the activity of this enzyme is responsible for the increased growth phenotypes and changes in labeling that they observe.

Based on the Reviewers' suggestions we performed these experiments. Herein, we took advantage of Y304A/G306E ETFDH mutant that impairs electron transfer from ETF and cannot substitute for the wild type (WT) gene function in ETFDH-deficient myoblasts [8]. We expressed WT and Y304A/G306E ETFDH mutant in ETFDH KO HCT116 colorectal cancer cells and confirmed that they are expressed to a comparable level (Supplementary Figure 6C). Re-expression of WT decreased proliferation, while suppressing mTORC1 signaling and increasing 4E-BP1 levels relative to control (vector infected) ETFDH KO EV HCT116 cells (Supplementary Figure 6D). In contrast, proliferation rates, mTORC1 signaling and 4E-BP1 levels remained largely unchanged upon Y304A/G306E ETFDH mutant expression in ETFDH KO HCT116 cells (Supplementary Figure 6D). Similarly, re-expression of WT ETFDH disrupted the bioenergetic phenotype associated with ETFDH loss, in contrast to re-expression of Y304A/G306E ETFDH mutant, which exhibited similar bioenergetic profiles as ETFDH KO control (Supplementary Figure 6E-F). Collectively these findings argue that the ETFDH activity is required for its tumor suppressive effects.

If nucleotide pool and labeling data are available, or can be obtained readily, this would significantly strengthen the tracing data already obtained.

We followed Reviewer's suggestion and measured nucleotide levels. This revealed that loss of ETFDH results in increase in steady-state nucleotide pools (Supplementary Figure 2K), consistent with increased aspartate labelling and accelerated tumor growth.

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<https://doi.org/10.7554/eLife.106587.2.sa0>